



Tracking the Evolution of the Loss and Damage Financing Architecture under the UNFCCC

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K R Hari Krishna

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Research Guide

Dr Kamal Kumar Murari

BS Analytics and Sustainability Studies

Labour Market Research Facility

School of Management and Labour Studies

Tata Institute of Social Sciences, Mumbai

DECLARATION

I, K R Hari Krishna, hereby declare that this dissertation entitled '**Tracking the Evolution of the Loss and Damage Financing Architecture under the UNFCCC**' as the outcome of my own study undertaken under the guidance of Dr Kamal Kumar Murari, Centre for Climate Change and Sustainability Studies, School of Habitat Studies, Tata Institute of Social Sciences, Mumbai. It has not previously formed the basis for the award of any degree, diploma, or certificate of this Institute or of any other institute or university. I have duly acknowledged all the sources used by me in the preparation of this dissertation.

A handwritten signature in blue ink, consisting of the initials 'K.R.H.K.' enclosed within a circular loop.

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K R Hari Krishna

CERTIFICATE

This is to certify that the dissertation entitled “**Tracking the Evolution of the Loss and Damage Financing Architecture under the UNFCCC**” is the record of the original work done by **Hari Krishna K R** under my guidance and supervision. The results of the research presented in this dissertation/thesis have not previously formed the basis for the award of any degree, diploma, or certificate of this Institute or any other institute or university.

Date:

Dr. Kamal Kumar Murari,

Assistant Professor,

Centre for Climate Change

and Sustainability Studies,

School of Habitat Studies

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LIST OF ABBREVIATIONS

Abbreviation	Expanded Form
AIIB	Asian Infrastructure Investment Bank
AOSIS	Alliance of Small Island States
AR6	Sixth Assessment Report (IPCC)
ARC	African Risk Capacity
BCR	Benefit-to-Cost Ratio
BIM	Barbados Implementation Modalities
C3S	Copernicus Climate Change Service
CBA	Cost-Benefit Analysis
CBDR-RC	Common but Differentiated Responsibilities and Respective Capabilities
CCRIF	Caribbean Catastrophe Risk Insurance Facility
CDA	Critical Discourse Analysis
CMA	Conference of the Parties serving as the Meeting of the Parties to the Paris Agreement
CO ₂	Carbon Dioxide
COP	Conference of the Parties
CP	Conference of the Parties (in decision numbering, e.g., 2/CP.19)
CPI	Climate Policy Initiative
ELD	Economic Loss and Damage
EM-DAT	Emergency Events Database
EMDE	Emerging Market and Developing Economies
EU	European Union
EUR	Euro (currency)
FRLD	Fund for Responding to Loss and Damage
G77	Group of 77 (developing countries coalition)
GBP	British Pound Sterling
GDP	Gross Domestic Product
GEF	Global Environment Facility

GHG	Greenhouse Gas
GNP	Gross National Product
IBRD	International Bank for Reconstruction and Development
INC	Intergovernmental Negotiating Committee
IPCC	Intergovernmental Panel on Climate Change
L&D	Loss and Damage
LDC	Least Developed Country
MDB	Multilateral Development Bank
NDC	Nationally Determined Contributions
NELD	Non-Economic Loss and Damage
NPV	Net Present Value
OECD	Organisation for Economic Co-operation and Development
SBI	Subsidiary Body for Implementation
SBSTA	Subsidiary Body for Scientific and Technological Advice
SCCF	Special Climate Change Fund
SIDS	Small Island Developing States
UAE	United Arab Emirates
UK	United Kingdom
UN	United Nations
UNDP	United Nations Development Programme
UNEP	United Nations Environment Programme
UNFCCC	United Nations Framework Convention on Climate Change
US	United States
USD	United States Dollar
WIM	Warsaw International Mechanism
WMO	World Meteorological Organization

ABSTRACT

Developing countries suffered \$320 billion in climate losses in 2024 and projections reach \$1.7 trillion annually by 2050. The international community has negotiated for three decades yet the only dedicated global mechanism held just \$828 million as of late 2025. This sum is less than 0.2 per cent of the annual need and the gap shows no sign of closing under the current design.

This study argues that this gap is structural and built into the legal architecture of the fund itself. Negotiators systematically removed liability language from every major UNFCCC agreement between 1991 and 2023. This dynamic is named destructive ambiguity, where textual vagueness enables consensus but destroys the capacity to generate predictable finance.

Drawing on Fairclough's critical discourse analysis, this study examined UNFCCC decision texts and negotiating documents from 1991 to 2024. Analysis revealed how liability-linked terms declined while cooperation language rose and how mandatory financial proposals were rejected or diluted before reaching a final text.

The findings confirm three things about the current financing architecture. First, exclusion of liability translated directly into clauses that forbid mandatory targets and leave contribution volumes entirely to donor discretion. Second, the free-rider problem has become a designed feature of a voluntary model without enforcement mechanisms or contribution formulas. Third, solidarity levies offer the most viable path to predictable revenue without triggering the liability red lines that developed nations have enforced since 1991.

The upcoming World Bank review is the immediate legal opportunity to redesign the fund's revenue architecture through COP decisions as no improvement in governance can close the gap between the billions needed. Thus, addressing loss and damage requires changing the legal basis for contributions and the clock is already running.

Keywords: loss and damage, UNFCCC, Fund for Responding to Loss and Damage, critical discourse analysis, destructive ambiguity, liability, climate finance, constructive ambiguity, fiscal authority, voluntary contributions

INTRODUCTION

Background

Global average temperatures have breached critical thresholds, destroying economic assets and human lives (Germanwatch e.V., 2025). With Earth already recording its hottest day on 22 July 2024, and with 2024 recording the highest average temperature on record, 1.5°C above pre-industrial levels, there is a necessity to either curb or adapt to the increase in temperature (C3S, 2025; WMO, 2024). These escalating impacts have driven demands for a dedicated financial architecture, the evolution of which this study traces

Anthropogenic greenhouse gas emissions drive this warming, altering the intensity, frequency, and duration of destructive climatic events. Under current policy pathways, global temperatures will rise by 2.6°C to 3.1°C by 2100 (UNFCCC, 2025b). These warming patterns trigger severe impacts through extreme weather events and slow-onset processes.

While extreme weather events involve sudden, discrete shocks such as tropical cyclones, flash floods, and heatwaves that destroy infrastructure, cause loss of life, and disrupt supply chains, slow-onset processes evolve gradually over years or decades without a clear start or end point (D. Coumou et al., 2024; Fischer & Knutti, 2015; Limsakul et al., 2024). However, rather than operating in isolation, these mechanisms compound and exacerbate each other (Limsakul et al., 2024). Rising sea levels, for example, raise the baseline water level, allowing storm surges from tropical cyclones to penetrate farther inland and cause disproportionate coastal flooding (Hegglin et al., 2022; Thomas et al., 2020).

The economic costs of these impacts are severe, unresolved by current adaptation efforts, and accelerating. Researchers project that residual climate-related losses and damage in developing countries will range from \$290 billion to \$580 billion annually by 2030, rising to \$1.132 trillion to \$1.741 trillion by 2050 (Bhandari et al., 2022; Markandya & González-Eguino, 2019). The global economy is already affected, with climate-related disasters causing \$320 billion in direct losses in 2024 (MunichRE, 2025).

These estimates are conservative as they measure only direct physical damage and miss indirect losses such as reduced labour productivity, supply chain failures, and unmonetised ecosystem destruction. This gap matters for Loss and Damage governance, as current financial and

institutional responses are designed around visible, short-term losses, leaving large parts of the actual economic burden unaccounted for and underfunded.

Historical data from EM-DAT, compiled by Delforge and team (2025), confirm the cumulative toll of these climate hazards. Between 01 January 2000 and 14 April 2026, the world experienced over 9,803 distinct extreme weather events, resulting in over 1,571,197 deaths globally, with direct, inflation-adjusted economic losses reaching nearly \$4.8 trillion. Storms caused the greatest economic destruction, accounting for 49.6 per cent of total losses at \$2.4 trillion, followed by floods, which accounted for 21 per cent at \$1.1 trillion. These figures show the scale and concentration of losses, reinforcing the need for governance frameworks that can address both cumulative and event-specific Loss and Damage.

These aggregate figures hide local vulnerabilities, as single high-impact events could obliterate the entire economic output of exposed nations. In 2017, Hurricane Maria struck Dominica, causing up to \$1.8 billion in damage, equivalent to 270 per cent of the nation's annual GDP (Germanwatch e.V., 2025). Similarly, Cyclone Pam devastated Vanuatu in 2015, inflicting damages equal to 64 per cent of its national GDP (Germanwatch e.V., 2025). While SIDS and LDCs are more susceptible to such hazards, larger developing economies also suffer massive proportional losses. In 2022, extreme rainfall caused flooding across one-third of Pakistan, affecting 33 million people, killing over 1,700, and causing nearly \$15 billion in direct damage, destroying roughly 10 per cent of the nation's GDP (Germanwatch e.V., 2025). These extreme financial shocks force vulnerable countries to take on risky foreign debt to secure immediate disaster relief, thereby reducing fiscal space and constraining long-term economic growth.

Three Pillars of Climate Action and Finance

Mitigation

The international community has responded to the rising economic and human costs of climate change through Mitigation and Adaptation in climate action and finance. **Mitigation targets the physical causes of climate change and requires reducing GHG emissions and enhancing carbon sinks** (UNEP, 2023c). Governments implement mitigation through energy transitions, energy efficiency improvements, and carbon markets to reduce emissions.

While the Paris Agreement aims to limit global warming to 1.5°C above pre-industrial levels, current NDCs project a temperature increase of 2.6°C to 3.1°C by 2100, indicating that mitigation efforts remain insufficient to stabilise the climate (UNFCCC, 2015a). This gap

between targets and outcomes means that significant climate impacts and associated losses are unavoidable, directly linking mitigation failure to the scale of Loss and Damage.

Adaptation

Adaptation involves structural and behavioural adjustments to reduce human and ecological vulnerability to expected climate impacts, including communities building sea walls to hold back rising tides, planting drought-resistant crops to secure food supplies, and installing early warning systems to evacuate populations before cyclones make landfall (UNEP, 2023b). While mitigation reduces future risks, adaptation protects existing assets from anticipated damage, preserving economic value and avoiding future costs (UNEP, 2023a). However, adaptation has limits, beyond which losses cannot be avoided.

Loss and Damage

Loss and Damage (L&D) refers to the actual or potential manifestation of adverse effects due to climate change that people cannot cope with or adapt to (Huber & Murray, 2024; McDonnell, 2023; Nand et al., 2024; UNFCCC Secretariat, 2012; Warner & van der Geest, 2013). L&D covers impacts from sudden, extreme weather events and gradual, slow-onset events, particularly in developing countries, which are more vulnerable to such impacts (UNFCCC, 2014). In this context, loss consists of effects that cannot be fixed easily, leading to irreversible changes, while damage refers to harmful impacts where restoration is possible but leads to high costs and burdens for the affected (Bhowmik et al., 2021; Doktycz & Abkowitz, 2019; Nand et al., 2024).

"Loss and Damage" could be considered a product of the limitations of the conventional pillars of climate action, i.e., mitigation and adaptation. L&D refers to the adverse effects of climate change, which are often inevitable and irreversible, and persist even if, or even in the absence of, successful implementation of the above two strategies (UNDP, 2023; UNEP, 2023d).

L&D thus represents “residual risk”, with impacts that are often irreversible and include economic and non-economic damages such as the loss of homes and livelihoods due to erosion, degradation of agricultural land, trauma from extreme weather events, and the loss of sacred sites due to sea-level rise (UNDP, 2023). These residual impacts also reveal the limits of adaptation, and understanding them helps determine the scale and urgency of mitigation required globally (Klein et al., 2014).

Types of L&D

Categorisation based on Valuation Methodology

Loss and Damage are generally categorised into two main types: economic and non-economic forms (Bahinipati, 2020; Nand et al., 2024; UNFCCC Secretariat, 2012).

Economic Loss and Damage (ELD) impacts resources, goods, and services. This does include tangible losses such as damage to crops, public and private property, reduced income, and loss of farm employment (Bahinipati, 2020; García-Portela, 2024). Other economic impacts include land turning to desert, loss of livestock, additional irrigation costs, and damage to existing infrastructure and other assets (Bahinipati, 2020; Boda et al., 2021).

Non-Economic Loss and Damage (NELD) covers impacts on life that do not have a direct market value or are not traded in markets, such as the loss of cultural heritage, traditional knowledge, and biodiversity (Bahinipati, 2020; Fezzigna et al., 2025; García-Portela, 2024; Nand et al., 2024; van Schie et al., 2024). Human-related NELD consists mainly of intangible and indirectly tangible costs such as loss of life, physical and mental health, mobility, conflicts over water, declining groundwater levels, and harm to forests or water quality (Bahinipati, 2020; Gewirtzman et al., 2018; Nand et al., 2024; van Schie et al., 2024). It can also include changes in productive land, emotional distress, the sacrifice of celebrations, temporary relocation, and permanent changes in one's profession (Bhowmik et al., 2021).

NELD is highly context-specific and often difficult to define, measure, or price (Boda et al., 2021; Nand et al., 2024; Scown et al., 2025; van Schie et al., 2024). NELD is also systematically overlooked in policies, making it a significant blind spot in many current assessments, as it is difficult to conceptualise, account for, and monetise (Bahinipati, 2020; Doktycz & Abkowitz, 2019; Huber & Murray, 2024).

Developed nations generally report higher absolute economic L&D (L&D) due to their extensive capital, infrastructure, and insured assets (Bahinipati, 2020; Boda et al., 2021; Boyd et al., 2021; Fezzigna et al., 2025). However, developing countries often face a greater economic L&D when measured as a percentage of their national income or GDP because their economies are less able to absorb economic shocks, and their livelihoods often depend heavily on natural capital (Bahinipati, 2020; Dorkenoo et al., 2022; Fezzigna et al., 2025; Martyr-Koller et al., 2021). For instance, a study conducted by Bahinipati (2020) in rural India found that farmers experienced high immediate tangible and intangible losses, highlighting the

disproportionate impact of climate-related disasters on more vulnerable economies and populations, where even minor absolute losses can represent a significant portion of their limited resources.

Categorisation based on Prevention and Risk Reduction

Loss and Damage can also be understood by considering whether effective climate action could have lessened their effects, to grasp the importance of taking meaningful steps to combat climate change and its consequences.

Firstly, **Avoidable effects** refer to the negative impacts that could be minimised or prevented through planned adaptation options through substantial mitigation and adaptation efforts (Bahinipati, 2020; Mechler et al., 2020).

Meanwhile, **Unavoided effects** result from missed opportunities for planned adaptation actions, but are constrained by technical, financial, or socio-economic constraints (Bahinipati, 2020; Bhowmik et al., 2021; Mechler & Deubelli, 2021; Warner & van der Geest, 2013).

Finally, **Unavoidable effects** refer to the impacts of climate change that cannot be stopped, even after trying to reduce or adapt to it and are often called "residual impacts", these occur when societies hit their limits to adaptation (Aleksandrova & Costella, 2021; Bahinipati, 2020; Bhowmik et al., 2021, 2021; Fekete & Sakdapolrak, 2014).

L&D and Adaptation Limits

The Intergovernmental Panel on Climate Change (IPCC) explains that these limits occur when a society can no longer achieve its objectives, protect its values, or avoid risks deemed intolerable, even with adaptation measures (Calvin et al., 2023). There are two main types of adaptation limits:

Hard limits mean no adaptation options are physically possible to avoid intolerable risks, and when crossed, any action will not be effective, leading to permanent harm (Calvin et al., 2023; Mechler & Deubelli, 2021; Roberts & Pelling, 2018).

Soft limits occur when adaptation options are unavailable due to a lack of funding, administrative or technical capacity, or other socio-economic barriers. However, development interventions can sometimes postpone or remove these soft limits (Calvin et al., 2023; Mechler et al., 2020; Mechler & Deubelli, 2021; Roberts & Pelling, 2018).

L&D goes beyond adaptation because it involves situations in which action becomes ineffective, leads to maladaptation through adverse outcomes, or is simply not possible due to concerns beyond adaptation's scope (Robinson et al., 2021; van Schie et al., 2024). Even if efforts are successful, some residual L&D is expected, and these often lead to permanent losses.

Importance of L&D

L&D has gained recognition as a distinct pillar alongside mitigation and adaptation in international climate policy frameworks such as the Warsaw International Mechanism (WIM) and Article 8 of the Paris Agreement, guided by the principle to “avert, minimise, and address” climate impacts (UNFCCC, 2014, 2016b). This reflects the growing acknowledgement that some climate impacts cannot be prevented or adapted to, particularly in vulnerable regions.

Climate impacts are increasing in frequency and severity, disproportionately affecting developing countries that have contributed the least to global emissions (Boyd et al., 2021; Calvin et al., 2023; Gewirtzman et al., 2018; Huber & Murray, 2024; McDonnell, 2023; Nand et al., 2024; van Schie et al., 2024). When losses are not measured or accounted for, especially non-economic impacts, the true scale of damage and recovery needs is underestimated. This leads to gaps in finance, data, and governance systems. A clearer understanding of L&D helps identify these gaps and supports the design of more effective, inclusive, and targeted responses for vulnerable populations (Bahinipati et al., 2024; Clarke et al., 2023).

The Financing Problem of L&D

The global climate finance architecture cannot manage permanent, unpriceable destruction. Multilateral development banks (MDBs), bilateral donors, and private investors supply the vast majority of international climate capital, and they strictly operate on a market-return logic, favouring projects that generate revenue or protect existing capital (CPI, 2023). Consequently, mitigation projects and adaptation infrastructure dominate the funding landscape, leaving little explicitly earmarked for L&D (CPI, 2023; UNFCCC, 2025b).

Mitigation attracts hundreds of billions of dollars annually because it creates profitable assets. In 2021 and 2022, global mitigation finance reached USD 1.15 trillion, of which the majority was funded by private capital through project-level market-rate debt and equity (CPI, 2023), producing measurable returns.

Adaptation finance operates on a different, but equally market-driven, logic by preserving the value of the targeted resource. These interventions aim to shield existing economic assets from projected climate hazards (CPI, 2023).

Economists, such as Mechler and team in 2019, justify these investments using CBA, NPV and BCR. For example, constructing a ring dike around a coastal city requires immense capital, but it prevents future flood damage to real estate and infrastructure. The avoided future cost functions as a tangible financial benefit. Because adaptation protects future economic output, development banks and governments can justify the upfront costs. When institutions do fund adaptation, they rely heavily on debt. Between 2016 and 2022, loans constituted the majority of climate financing provided by MDBs (CPI, 2023). Forcing highly vulnerable nations to borrow to rebuild after unavoidable disasters worsens their existing debts, strips them of fiscal power, and creates a vicious cycle of ruin.

L&D finance, on the other hand, is incompatible with both the revenue-generation logic of mitigation and the value-preservation logic of adaptation (Boda et al., 2021; CPI, 2023). Rebuilding a hospital destroyed by a Category 5 hurricane does not generate new economic growth; such interventions restore assets rather than expand capital or yield financial returns. Consequently, private investors and commercial banks have little incentive to fund L&D recovery.

Since L&D is the realisation of residual risks, it creates a distinct fiscal challenge as it generates no revenue, and conventional financial instruments are not designed to address such losses. Equity investments require an underlying asset that appreciates or generates dividends. A submerged coastal village or a salinised agricultural plain offers no such asset, leaving loans as the primary conventional tool. However, deploying loans for L&D creates a severe structural injustice. Multilateral development banks and bilateral donors currently provide 63 per cent of adaptation finance in the form of debt (UNEP, 2023). Applying debt to L&D forces highly vulnerable nations to borrow to recover from disasters they did not cause.

For residual risks, the system relies on insurance mechanisms such as catastrophe bonds and risk-pooling facilities like the African Risk Capacity (ARC) and the Caribbean Catastrophe Risk Insurance Facility (CCRIF) (Adeniyi, 2025; Gewirtzman et al., 2018; UNDP, 2023). Catastrophe bonds transfer disaster risk to capital markets, offering investors high yields while providing governments with liquidity when predefined triggers are met.

However, this reliance on commercial insurance fails to account for the physical realities of L&D. While insurance is effective for rare, sudden events such as storms, it is ineffective for slow-onset processes such as sea-level rise or desertification, which are certain to occur (Gewirtzman et al., 2018; Robinson et al., 2021; Verchick, 2018). In 2015, severe floods in Malawi caused US \$366 million in losses but triggered an insurance payout of only US \$8.1 million, highlighting the mismatch between coverage and actual damages (Adeniyi, 2025).

When extreme weather destroys a nation's infrastructure, its Gross Domestic Product (GDP) contracts sharply, requiring governments to finance recovery through sovereign borrowing (UNEP, 2023; Adeniyi, 2025). To fund immediate relief and reconstruction, governments issue new debt, often at high interest rates, which reduces fiscal space. As a result, domestic revenues are diverted away from essential sectors such as education, healthcare, and future climate adaptation towards debt servicing. Over time, repeated climate shocks can trap countries in cycles of debt distress, constraining long-term economic development. Conventional debt instruments thus do not resolve L&D; they convert physical losses into prolonged financial liabilities.

The financing problem intensifies when addressing NELD. Since it encompasses intangible and abstract losses, it defies conventional economic models because these losses cannot be reduced to a common measure (Mechler et al., 2019). Financial markets require assets to be expressed in a uniform metric, typically currency, to enable valuation and exchange. However, no financial instrument can rebuild lost cultural identity, restore an extinct language, or compensate for the permanent submergence of sovereign territory (Allen et al., 2024; Mechler et al., 2019). The value of these losses lies in social and ecological relationships, making traditional Cost-Benefit Analysis inadequate.

Attempting to monetise NELD also raises ethical concerns, as valuing human life or culture based on economic productivity implies unequal worth across countries and populations. Additionally, such models exclude NELD because it lacks market prices, underestimating the true scale of climate risk. As a result, current market-based mechanisms treat these losses as externalities, rendering them unfundable through standard financial mechanisms.

The operational logic finance shows that L&D requires a fundamentally different financial architecture. Because L&D interventions address irreversible losses, generate no financial returns, and include non-economic impacts that cannot be monetised, they cannot be financed through conventional instruments.

A suitable model must therefore rely primarily on non-repayable, grant-based finance, addressing losses without imposing additional debt burdens on vulnerable countries.

The Research Problem

This particular financial need drives the political discourse on responsibility, but different parties have taken different positions over time. Developing countries insist on a liability system that mandates funding. Developed countries, on the other hand, prefer to offer voluntary support grounded in "cooperation" and "solidarity," ensuring that L&D finance does not entail legal liability or a right to compensation. The problem of L&D finance is thus how to secure adequate, reliable funding for irreversible harm without the legal admission of guilt that developed nations refuse to accept (Calliari et al., 2018; Yvonne Karimi-Schmidt, 2020).

This tension is evident in the Paris Agreement as shown by **Decision 1/CP.21, which explicitly states to “not involve or provide a basis for any liability or compensation” (UNFCCC, 2015b, 2016a)**. With voluntary financial support, funding cycles are unpredictable and depend on fiscal conditions and the political will of donor nations, rather than being tied to historical responsibility or the scale of the damage inflicted on the vulnerable (Åberg & Jeffs, 2022; van Schie et al., 2024). With no justification for levies or taxes being imposed on historical emitters, this further pushes L&D and Climate Finance in general towards traditional grant-based aid (Pill, 2022; Wewerinke-Singh & Salili, 2020).

The operationalisation of the FRLD at COP28 mirrors this shift from obligation to cooperation with the fund, hosted by the World Bank rather than being operated as an independent entity with assessment powers, along with donor preferences for voluntary contributions and stringent fiduciary oversight, rather than the other disbursement methods favoured by developing nations (Schalatek, 2025).

The research problem thus centres on the current L&D financing architecture, which was intended to provide substantial, predictable support to address the impacts of climate change. However, the framework does not allow for liability and compensation, thereby preventing a direct connection between responsibility and financial support. This makes the mobilisation of revenues dependent on voluntary rather than mandatory means. The institutional question, therefore, revolves around the system's ability to generate the necessary revenue.

This study thus seeks to establish whether the shift from a liability-based to a cooperation-based system has affected the authority of the L&D financing framework to mobilise revenues, particularly the FRLD's capacity to mobilise funding.

Normative Basis	Financial Logic	Contribution Structure	Fiscal Authority	Revenue Outcome
Liability and Compensation	Corrective Justice	Mandatory Contributions	Strong Fiscal Authority	Predictable and scalable revenue mobilisation
Cooperation and Solidarity	Distributive Justice	Voluntary Contributions	Weak Fiscal Authority	Unpredictable and politically contingent financing

Table 1.1: Author's conceptualisation based on Wallimann-Helmer et al. (2018), Mechler et al. (2019), Verchick (2018), and Calliari (2018).

Aim

The study tracks the development of L&D within the UNFCCC and assesses the extent to which the shift from 'liability and compensation' to 'cooperation and facilitation' structurally limits the Fund for Loss and Damage (FRLD) 's potential to mobilise revenue.

Objectives

1. Building a timeline of primary sources of L&D policy documents, decisions, draft negotiating texts, and country submissions from the initial 1991 AOSIS proposal to the 2024 operationalisation of the FRLD,
2. Analysing primary sources of L&D in the UNFCCC from two periods: pre-Paris (1991-2015) and post-Paris (2016-2024),
3. Creating a map of country-level negotiating positions to assess which countries shifted from mandating compensation to embracing solidarity contributions,
4. Analysing the range of financial instruments proposed in the UNFCCC negotiations vs. those included in the final form of the FRLD, and the amount pledged in each case.
5. Evaluate notable clauses in the FRLD Governing Instrument to determine the legal limits to the revenue authority's power on mandatory contributions.

Research Questions

Does the shift in the UNFCCC texts from a focus on liability to one on cooperation structurally limit the FRLD's authority to enforce mandatory revenue measures?

Sub-questions:

1. How does the use of words like "compensation," "restitution," and "insurance pool" change in the negotiation texts across the time periods?
2. Which country groups altered their positions on financial obligation in the shift from the Warsaw International Mechanism (WIM) to the FRLD?
3. Which financial measures were eliminated in the final decisions based on liability concerns?

4. Does the FRLD Governing Instrument provide the FRLD with legal authority to enforce mandatory capitalisation targets or levies?
5. Which legal and financial limitations on "innovative sources," such as taxation, can be found in the legal *modus operandi* of the FRLD?

Scope

The scope of the study is the institutional and textual evolution of L&D finance under the UNFCCC. The study will cover various official decision texts, draft negotiating texts, country submissions, and governing documents from 1991 to the present. The study will also cover various proposed and adopted financial instruments, as well as publicly disclosed pledges and contributions.

The study will specifically focus on the content analysis of the definition of financial authority as enunciated under the law. The study will also focus on the evolution of terms and changes in the framing of authority and revenue across the different negotiation periods.

Limitations of the Study

The FRLD was created in late 2023 and is yet to be operationalised as of May 2026. This leaves us with no ex-post disbursement data. As a result, the analysis remains limited to the ex-ante institutional design outlined in the Governing Instrument. Additionally, it is important to note that the datasets underreport losses in low-income countries and do not include NELD data, further limiting the analysis. Given that the effectiveness of the voluntary pledge model has not yet been tested, there are concerns regarding potential “delivery gaps” that may arise.

Additionally, constructive ambiguity in UNFCCC texts presents a coding challenge. Negotiators often use vague language to build consensus, allowing diverse interpretations of clauses to coexist. For instance, the phrase “addressing loss and damage” remains intentionally undefined to prevent stalling discussions. This ambiguity makes it difficult to categorise the text, requiring interpretive judgment that the text doesn’t clarify.

The institutional arrangement for the FRLD is also temporary and will undergo a formal review after four years. Future COP decisions may change the Fund's legal structure, governance authority, or hosting institution, potentially altering the design features analysed in this study. This assessment focuses on the FRLD’s current institutional architecture as it begins operations, without predicting specific future changes, since the governance arrangement is subject to revision.

Additionally, the study emphasises the formal negotiation regime of the UNFCCC and does not systematically explore parallel processes, such as the bilateral climate finance arrangements, which may influence FRLD outcomes in ways not fully captured here.

Academic Contribution

This research adds to the literature on climate governance by verifying the connection between normative theory and fiscal authority. Current literature either analyses legal texts or estimates financial gaps. This research bridges the two, showing that the removal of "liability" is not just a legal defence mechanism for developed countries but also a design choice that actively divests international institutions of fiscal authority.

This research adds to the literature on institutional design by analysing the effect of "constructive ambiguity" on "destructive ambiguity." It thus offers a new model for analysing the impact of soft law on hard financial constraints.

Policy Relevance

The research findings will reveal the exact structural flaws in the FRLD that make it susceptible to the "empty shell syndrome." In fact, the research proves that the current "voluntary pledge" model is a direct function of the normative exclusion of liability and thus the FRLD lacks the necessary statutory tools to raise the projected USD 400 billion annually by 2030.

The research identifies that, to attain financial adequacy, it is necessary to look beyond the current "cooperative" text. It implies that in the next round of negotiations, there is a need to reintroduce some form of mandatory obligations, possibly through creative solidarity levies or taxes, that are not within the liability "red lines" of the UNFCCC, to ensure predictability that is lacking in the current FRLD.

LITERATURE REVIEW

Normative Frameworks in L&D

Liability Frameworks

As the analysis in Chapter 1 showed, the failure of current climate finance mechanisms stems from treating climate disaster relief as humanitarian aid, effectively transferring the physical and financial consequences of historical high-emission economic growth from industrialised nations to the Global South. This is the fundamental injustice that the liability model, which requires the causes of damage to pay for the repair, is designed to address.

The liability model rests on the philosophical foundation of corrective justice. Corrective justice, as defined by Wallimann-Helmer et al. (2018), establishes a direct moral and legal relationship between the actor who causes harm and the victim. When an identifiable entity inflicts an injury through wrongful conduct, that entity incurs a strict duty to make the victim whole, with the perpetrator required to restore the victim to the **status quo ante**, i.e., the exact position they would have occupied had the harm never occurred. Applied to climate change, this logic dictates that the historical emissions of industrialised nations caused the climate shifts now devastating developing countries. Therefore, these heavy emitters bear a strict obligation to fund the resulting L&D. In 1991 (UNFCCC, 1991b, 1991a), and later in 2008, through **Decision 1/CP.13** (UNFCCC, 2008), the Alliance of Small Island States (AOSIS) asserted that compensation for climate impacts is a legal entitlement based on historical responsibility. These proposals included mechanisms in which mandatory contributions from industrialised nations would fund compensation for sea-level rise and extreme weather, directly linking payments to emissions levels and economic capability.

International law codifies this through the doctrine of state responsibility and the **no-harm rule**, requiring states to prevent activities within their jurisdictions from causing significant transboundary harm (Mechler et al., 2019; Simlinger & Mayer, 2018). This principal traces back to the *Trail Smelter Case (United States v. Canada)* in 1941, which established that no state has the right to use its territory to cause severe injury to another state. If a state breaches this obligation by emitting excessive greenhouse gases, it commits an internationally wrongful act. Under the standards established by the *Factory at Chorzów Case* in 1928, full reparation

requires the state to wipe out all consequences of the illegal act, typically via financial compensation when physical restitution is impossible. Consequently, climate-vulnerable states argue that the failure of industrialised nations to curb emissions breaches the no-harm rule, triggering a legal duty to compensate for the resulting L&D.

Domestic environmental frameworks popularised this mechanism as the **polluter pays** principle, which advocates internalising the social and environmental costs of their actions rather than passing them on to the public (Ohdedar, 2016). In the international arena, this principle exerts immense political force because it assigns clear moral culpability to high-emitting nations (Kumar, 2024; Siegele, 2023). Theorists argue that it faces limitations when applied to historical emissions, as early emitters were often unaware of the harm they caused and the responsible actors are no longer identifiable (García-Portela, 2023). To resolve this, some scholars propose the **Beneficiary Pays** Principle, which shifts the focus from the act of polluting to the economic benefits it generates (Wallimann-Helmer et al., 2018). Despite this theoretical refinement, the polluter pays principle remains the dominant political narrative because it directly aligns with the corrective justice demand to hold wrongdoers accountable.

However, enforcing strict liability for global climate damage faces a significant practical barrier, as corrective justice requires a direct causal link between a specific wrongful act and a specific injury. Climate change operates through cumulative emissions over decades, making it difficult to attribute a single extreme weather event to a single nation's emissions (Mechler et al., 2019; Olsson, 2022; Otto, 2023). Because extreme event attribution relies on probabilistic methods (estimating changes in likelihood and intensity rather than establishing direct causation) it does not align neatly with legal standards that require clear, linear causation. As a result, strict liability models face a structural limitation under conditions of diffuse, cumulative causation, rather than outright failure. Industrialised nations exploit this gap to contest their obligation to pay for specific damages, keeping the liability model politically salient while leaving it legally unresolved.

Cooperative Model

To resolve this political deadlock over liabilities, international climate frameworks increasingly rely on the **cooperation (or solidarity) model**, grounded in **distributive justice**. Where corrective justice looks backwards to identify who caused harm and assigns liability, distributive justice looks forward and asks how the community should allocate burdens fairly to correct an unfair baseline distribution of resources and risks. Under this model, severe

climate impacts become undeserved harms due to geographic bad luck, rather than the direct result of wrongful emissions by specific states. Consequently, support becomes a humanitarian response to severe misfortune rather than a legal obligation to repair an inflicted injury.

This shift in philosophical framing alters the entire basis of financial responsibility. Wallimann-Helmer et al. (2018) show that when L&D is viewed as an undeserved harm requiring redistribution, the focus shifts from historical emissions and legal culpability to the ability-to-pay principle, under which parties with the most resources and excess capacity must contribute the most to the common endeavour, regardless of their historical carbon footprint. This relies on voluntary, progressive contributions driven by current wealth and capacity while abandoning mandatory, assessed contributions tied to a nation's specific emissions record. It thus shifts from rectifying a past wrong to evening out a current unfair disadvantage. As a result, assistance often focuses on the recipient's urgent needs rather than the funder's fault.

Verchick (2018) and Mechler et al. (2019) show that separating financial support from causation removes the legal mechanism that binds specific polluters to specific damages, moving away from accountability and towards charity. The funding mechanism degrades from restitution, triggered by a legal responsibility, to voluntary aid, dependent entirely on donor willingness and current fiscal capacity. Because donors contribute out of solidarity rather than legal obligation, they retain the power to dictate the terms, timing, and scale of their financial support.

Pill (2022) notes that solidarity offers a highly practical compromise in global negotiations because it recognises the impacts felt by developing countries without assigning fault, thereby avoiding the legal ramifications and endless compensation claims that stall negotiations. Schinko et al. (2018) add that a distributive justice perspective is necessary in the short- to medium-term because attributing specific losses to specific anthropogenic emissions remains scientifically and legally complex. A solidarity fund appeals directly to human morale and expresses compassion, sidestepping the need for international courts or admissions of guilt. However, treating L&D finance purely as humanitarian aid risks double-counting development funds and allows developed nations to deflect their historical responsibility.

Wallimann-Helmer et al. (2018) defend the distributive justice approach on the grounds of non-ideal theory. Because proving direct causal attribution for every climate disaster remains complex, and enforcing liability remains politically deadlocked, demanding strict corrective justice leaves many victims with nothing. Verchick (2018), drawing on Judith Shklar's work,

argues that injustice flows not only from harmful acts but also from failures to act. Therefore, wealthier nations owe a duty to act simply because they have the means to prevent or relieve suffering, regardless of the origin of the injury. A distributive approach ensures that communities facing climate threats receive assistance based on their urgent needs. Furthermore, a solidarity framework captures all L&D, providing support regardless of whether a specific disaster stems directly from anthropogenic emissions or natural climate variability.

Wallimann-Helmer (2023) argues that L&D governance under a forward-looking model requires providing vulnerable communities with a sufficient threshold of capabilities to achieve climate resilience. If communities lack the financial and technological resources to recover from extreme weather events, they are entitled to support. This requires comprehensive risk management, early warning systems, and substantial capacity building. Consequently, ensuring a fair baseline distribution of resilience capabilities may require financial transfers as large as those demanded under strict liability.

Ambiguities and Coexisting Frameworks in Treaty Text

Multilateral climate treaties require all parties to be satisfied for any policy or decision to pass, yet parties may hold mutually exclusive positions on decisions to protect their own interests. To break these deadlocks, diplomats rely on constructive ambiguity. As Vanhala (2023) observes, negotiators use deliberately vague language that allows parties with fundamentally incompatible goals to interpret the text in accordance with their own interests.

The treaties in L&D Finance intentionally leave the phrase "addressing L&D" undefined. This ambiguity is seen with developing nations interpreting "addressing" as the first step towards a formal liability and compensation framework while the developed nations read the same word as a mandate for technical risk management, insurance schemes, and voluntary aid.

Calliari (2018) notes that vague treaty language systematically favours the party with less to concede. A liability model requires affirmative legal commitments, while the cooperation model requires only the absence of such binding text. Therefore, when negotiators use ambiguous language to bypass a definitive choice, the institutional architecture automatically defaults to voluntary cooperation. By refusing to assign responsibility, the text strips the framework of enforcement power. The deliberate omission of liability language constitutes a choice that shields developed nations from mandatory financial contributions.

This study terms this dynamic “destructive ambiguity”, a condition in which ambiguity enables institutional agreement but simultaneously undermines the framework’s capacity to generate enforceable obligations and adequate finance. This ambiguity allows treaties to exist also prevents them from raising capital, laying the foundation for **this form of destructive ambiguity.**

The Political Economy of Climate Finance

Incompatibility with L&D

As Boda et al. (2021) establish, the core definition of L&D, addressing assets, resources, and livelihoods that are irreversibly destroyed, is completely antithetical to the logic of private capital. Private investors require a yield-generating asset or collateral to secure their investments, but neither an inundated coastal village nor a salinised agricultural plain provides either (AIIB et al., 2025; Fritsch et al., 2025). Because L&D interventions simply restore basic survival conditions rather than expanding capital, they cannot generate the revenue streams that commercial banks and private investors demand.

NELD makes private financing even more unworkable as markets lack any pricing mechanism to value lost cultural heritage, indigenous knowledge, or human lives. Because financial instruments demand a uniform metric of exchange, they cannot process or fund the recovery of incommensurable losses. Attempting to force NELD into market-based frameworks distorts the true scale of the harm and leaves the most profound climate impacts entirely unfunded.

This becomes even more obvious when contrasted with the other pillars of climate action. While the mitigation and adaptation sectors successfully attract private capital due to their alignment with market incentives, L&D addresses the residual risks that materialise when adaptation fails, yielding a zero financial return.

Consequently, private capital will not bridge the L&D funding gap. The OECD (2021) confirms that private finance mobilisation for climate action remains highly uneven, flowing primarily to middle-income countries with investment-grade ratings. Private actors consistently refuse to finance projects that deliver public goods without commercial returns, either by avoiding high-risk contexts or by demanding guarantees, high returns, or government backing that vulnerable countries cannot provide. This shows that voluntary, market-based finance cannot address L&D, which does not generate profits and is concentrated in high-risk regions.

Concessional Finance and L&D

Multilateral development banks channel the majority of their climate finance through concessional loans (CPI, 2023). However, as Ike Uri et al. (2024) observe, using debt to address L&D forces vulnerable countries to underwrite the cost of climate disasters they did not cause. If a cyclone destroys a small island state's agricultural sector, demanding that the state borrow money to rebuild imposes a direct financial penalty on the victim. Adeniyi (2025) confirms that this "debt-on-disaster" approach is both unjust and destructive, as it increases debt-to-GDP ratios in countries already facing severe fiscal stress, diverting scarce public funds away from healthcare, education, and future climate adaptation simply to service foreign debt.

Policymakers frequently promote risk transfer mechanisms, such as catastrophe insurance, as the primary alternative to debt, only for debt to fail again. Insurance protects against low-probability, high-impact events that are random and unforeseen. Robinson et al. (2021) and Verchick (2018) demonstrate that this logic collapses when applied to slow-onset events such as sea-level rise, desertification, and the salinisation of freshwater. If a loss is certain, the required premium equals the payout, rendering the insurance mechanism entirely useless for long-term climate impacts.

Furthermore, insurance requires quantifiable damage assessments or specific parametric triggers to release payouts. Gewirtzman et al. (2018) establish that neither method works for NELD. Financial markets lack the mechanisms to price the permanent loss of a cultural heritage site, the erasure of indigenous knowledge, or the destruction of a traditional fishing ground. Because insurance demands a strict monetary metric for exchange, it inherently excludes the most profound human and ecological losses from its frameworks.

Free-Rider Problem in Voluntary Contribution Systems

International climate finance relies heavily on voluntary contributions, creating a severe collective action problem. Without binding legal obligations, the global institutional architecture cannot compel any individual state to contribute capital proportionally to its historical emissions or economic capacity. Consequently, every donor has an incentive to withhold funding and wait for other nations to bear the cost, thereby allowing targets to fail.

Åberg and Jeffs (2022) document that major contributors frequently make high-profile pledges to institutions like the Green Climate Fund but deliver only partial disbursements over time. Van Schie et al. (2024) show that these delivery gaps are structurally inherent to voluntary

financing models. They do not merely result from political bad faith. When governments face tight budgets, they often ignore international commitments, as there are no penalties for backing out.

If the FRLD operates voluntarily, the free-rider problem is not a peripheral operational risk but a consequence of a voluntary system, which explains exactly why the normative shift from liability to cooperation matters fiscally. By adopting a voluntary model, the UNFCCC institutionalises delivery gaps, ensuring that the architecture will systematically underfund the permanent climate losses it was created to resolve.

Documentation Bounding Institutional Authority

International organisations have no power of their own. As legal experts show, a body's powers and constraints derive exclusively from the multilateral agreements that established it. If a treaty provides that financial contributions are voluntary, the resulting body cannot legally require mandatory funding, compel countries to share costs, or punish states that refuse to pay. In the same way, if the treaty text clearly rules out liability, the body cannot create financial tools based on assessing fault or demanding repayment. The law's text sets the financial facts.

This legal boundary defines what the Fund for Responding to Loss and Damage (FRLD) can do. The FRLD manages the money side of the United Nations Framework Convention on Climate Change (UNFCCC) and the Paris Agreement. The Decision 1/CP.21 explicitly states to “not involve or provide a basis for any liability or compensation” (UNFCCC, 2016). Some, like Gsottbauer et al. (2018) and Wallimann-Helmer et al. (2018), argue that it blocks all compensation claims under the UNFCCC, rendering it unable to deliver climate justice. Others suggest it only limits Article 8 and does not eliminate general rights under international law or future mechanisms for allocating responsibility. Despite these interpretations, the practical outcome within the UNFCCC has been a shift from the “polluter pays principle” to a “victim pays” or “charity” approach.

This statement, however, has more than merely declaratory value and removes the juridical link between causation and compensation, disconnecting from the law on state responsibility. By excluding liability, the UNFCCC framework removed the legal mechanism for vulnerable nations to demand compensation for climate change-related losses, rendering financial support a discretionary political act rather than a binding legal duty (Simlinger & Mayer, 2018).

Because the foundational text removes liability, the FRLD cannot legally calculate specific damages owed by high-emitting states, nor can it compel those states to pay. Furthermore, the COP28 decision establishing the FRLD confirms that contributions remain entirely voluntary.

Literature Gaps Identified

Lack of Long-Term Analysis

The current discussion about L&D focuses on analysing the language used in negotiations, highlighting the divide between rich and poor countries, and examining how the idea of "constructive ambiguity" has changed over time. However, no analysis of all texts on L&D from 1991 to 2024 has been conducted; most research so far has focused only on IPCC reports, not on negotiation texts, leading to a lack of concrete evidence showing how terms related to liability have decreased while terms about cooperation have increased over the last 30 years of talks.

Missing Record of Financial Instruments

Research on financing L&D has concentrated mainly on potential future financial instruments, such as taxes on air travel or bunker fuels. While some historical accounts mention the initial proposal for a 1991 insurance pool, there has not been a comprehensive overview of the financial instruments proposed and then rejected throughout the negotiation process. The current literature suggests avenues for financing L&D, but fails to outline what was actually turned down and the reasons behind these rejections.

Limited Fiscal Authority

There has been insufficient exploration of how excluding "liability" affects the FRLD fiscal authority in collecting contributions or raising funds. These areas have typically been analysed separately rather than as interconnected issues.

METHODOLOGY

Research Philosophy

This study cannot use a positivist framework. Positivism assumes an objective reality independent of the observer, one that can be measured, tested, and verified. The FRLD's fiscal authority is a legal arrangement produced entirely by negotiated text, with its meaning and power determined by the words.

This study adopts an interpretivist epistemology grounded in a constructivist ontology. Constructivism holds that social actors create institutional reality through interaction, negotiation, and text production, rather than discovering a pre-existing one. For the purposes of this study, this means that the meaning of L&D is not fixed but actively constructed through competing textual interpretations in negotiations. In the UNFCCC context, "L&D" lacks a fixed natural meaning, as stated in decisions. When developing nations propose "compensation" and developed nations require "assistance," these are not stylistic preferences but are bids to construct an institution with fundamentally different legal powers.

A purely quantitative approach, counting pledges and calculating the funding gap, can describe the outcome but cannot explain it. Explaining why the FRLD was designed without mandatory revenue authority requires reading the texts that authorised it, the texts that preceded it, and the negotiating record that transformed the former into the latter.

Research Design

The central puzzle this paper addresses is how a concept as contested as 'loss and damage' acquired institutional form under the UNFCCC despite the lack of an agreed definition. Answering that question requires a method sensitive to language as a site of political contest, not simply a medium for communicating pre-existing positions. The writing of the texts (decisions, submissions, statements) produced by the UNFCCC negotiation process did not neutrally record political outcomes; rather, it was through the writing itself that the contentious issues surrounding the definition, scope, and responsibility for loss and damage were resolved. This shows how a focus on language as a mere vehicle for pre-existing interests overlooks the very process of interest formation and dispute.

Critical Discourse Analysis is well-suited to this context because UNFCCC negotiations are structurally unequal, with small island states and least-developed countries facing

industrialised blocs possessing vastly greater economic weight, legal capacity, and institutional access. Nevertheless, the outcomes, including the Warsaw International Mechanism and Article 8 of the Paris Agreement, do not straightforwardly reflect this power asymmetry. Understanding how parties shaped the outcome necessitates close attention to the relationship between language and power, which is the core concern of critical discourse studies.

Among all CDA methodologies, this paper builds on Fairclough's (1992, 2003) framework rather than, say, the discourse-historical approach of Wodak or the socio-cognitive model of van Dijk. These approaches are related, but they do not ask the same question. Fairclough is used here because it helps trace how wording in treaty text constructs institutional meaning over time. This CDA component helps answer whether parties frame L&D as a legal wrong requiring a remedy or as a shared misfortune requiring cooperation. What does the shift from "shall compensate" to "should assist" signal about institutional intent? How does removing a word from an operative clause strip an institution of a power it would otherwise hold?

The keyword discourse component systematically shows, across the full corpus, that liability-connected terms declined in discourse over time while cooperation-connected terms rose, thereby strengthening the causal argument. This design is **explanatory** and **descriptive** as it seeks to compare pledges, trace a historical narrative and survey actors.

Theoretical Framework

The theoretical framework draws on Fairclough's (1995, 2003) three-level model of CDA. At the textual level, the study examines specific lexical and grammatical choices, including the substitution of binding verbs (shall, must) with discretionary language (should, encourages), and the replacement of liability-connoting nouns (compensation, restitution) with solidarity-framing terms (assistance, voluntary support).

At the discursive level, the analysis examines how these texts were produced and consumed within the negotiation process, focusing on the specific junctures at which developing-nation proposals for strong legal obligation were diluted to secure consensus. At the social practice level, the study assesses the real-world institutional consequences of these textual shifts, in particular, whether the systematic removal of liability language has legally foreclosed the FRLD's authority to enforce mandatory financial contributions.

Data Sources

The dataset includes three different types of documents created within the UNFCCC framework from 1991 to 2024.

1. The Final adopted COP and CMA Decisions are the official records of the agreements reached, particularly those related to the Warsaw International Mechanism (WIM) and the Fund Mechanism for Loss and Damage (FRLD).
2. Country and Coalition Submissions to the Subsidiary Bodies (SBI/SBSTA) and the Transitional Committee reflect the positions of key groups, including AOSIS, the LDC Group, European Union, and Umbrella Group.
3. Meeting and Operational Documents of the Fund for responding to Loss and Damage.

The documents were chosen based on two main criteria. First, they must include the phrase “L&D,” which refers explicitly to the UNFCCC policy rather than just harm. Second, texts without this phrase were considered only if they discussed financial tools to address irreversible climate impacts, using terms such as “compensation,” “liability,” “insurance pools,” “risk transfer,” or “solidarity.” The documents that only mentioned “adaptation finance” without recognising the inevitable losses that cannot be adapted to were excluded.

Unit of	What is Coded	Purpose
Textual Segment	Paragraph or clause containing target keywords; verb modality (shall / should); operative vs preamble context	Qualitative discourse assessment
Financial Pledge	Individual commitment in USD. Pledged/committed/disbursed. source type (sovereign/philanthropic/innovative)	Quantitative financial analysis; delivery gap identification
Negotiating Position	Country or bloc submission; coded liability-supporting, cooperative-only, or neutral/ambiguous	Political analysis; tracking of demand migration

Table 3.1: Details on the unit of analysis and their purpose

Periodisation

The negotiations divide at 2015 with the pre-Paris period (1991-2015) covering the emergence of the L&D concept under the UNFCCC and Kyoto Protocol, leading up to the Paris Agreement. The post-Paris period (2016-2024) covers operationalisation with Article 8 work programmes, the Santiago Network, the Glasgow Dialogue, the Transitional Committee process, and the establishment of the FRLD. The 2015 dividing line is not arbitrary, as Decision 1/CP.21 formally excluded liability and compensation from the FRLD's legal mandate. This will serve as a point of reference for analysis.

Analytical Framework

This analysis uses a five-part causal framework to understand individual documents and compare different time periods. It suggests that how climate change is framed, either as a wrongful act needing legal responsibility or as a shared problem requiring cooperation, affects the legal language in treaty texts. This language shapes the institutional setup and influences the types of funding methods available, such as mandatory fees or voluntary grants. These factors ultimately impact how predictable and sufficient financial support is for affected countries.

By looking at this framework over time, we can see how early decisions, like the 2015 exclusion of liability in a key decision, have influenced later developments, including the financing design at COP27 and COP28. The analysis also identifies key moments when the focus shifted from obligation to cooperation, changing how funding works. Additionally, it examines the gap between high-level ethical principles and practical financial mechanisms, revealing instances in which ineffective funding mechanisms weaken strong ideas.

Analytical Techniques

1. **Keyword Trends:** This method involves examining the discourse of liability-related terms compared to cooperative terms over specified time periods. The analysis not only examines the term but also explores its context, distinguishing between clause types.
2. **Thematic Clustering:** This approach categorises text into overarching themes based on the concepts it contains. It helps illustrate shifts in framing over time, facilitating an understanding of how certain issues are perceived and articulated.
3. **Instrument Evolution Table:** This method classifies financial proposals by source and obligation type. Examining various financial mechanisms enables a comparison between proposed and adopted instruments, providing insights into evolving approaches.
4. **Country-Country Matrix.** A structured matrix maps AOSIS, the LDC Group, the European Union, and the Umbrella Group across periods, tracking fund flows among countries and entities.

FINDINGS AND ANALYSIS

Historical Evolution of L&D Governance

The 1991 AOSIS Proposal

AOSIS has been calling for compensation since the early 1990s. During the negotiations of the UNFCCC, the group proposed the establishment of an international insurance scheme explaining how “[t]he resources of the insurance pool should be used to compensate the most vulnerable small island and low-lying coastal developing countries for L&D resulting from sea level rise” (Intergovernmental Negotiating Committee, 1991; UNFCCC, 1991b). It relied entirely on assessed, mandatory contributions from industrialised nations and calculated these exact obligations using a strict formula: nations paid 50% based on their gross national product, representing their economic capacity, and 50% based on their relative share of global carbon dioxide emissions, representing their historical culpability.

The mechanism consisted of three connected parts: (a) an Insurance Element; (b) a Remediation/Compensation Element; and (c) a Risk Management Element. The Compensation Element was considered essential for addressing the remaining L&D caused by gradual processes such as rising sea levels, warming land and ocean surface temperatures, and ocean acidification.

Notably, the proposal lists State Responsibility among its foundational concepts, alongside the Polluter Pays Principle, common but differentiated responsibilities and respective capabilities (CBDR-RC), the precautionary principle, principles of fairness and intergenerational fairness, and international unity. The inclusion of State Responsibility is fundamentally linked to the idea of compensation, which, is a form of redress anticipated when State Responsibility arises from wrongful actions. Furthermore, the pool was one of the first to advocate for direct compensation for permanent loss, openly acknowledging that mitigation and adaptation alone cannot reverse climate impacts.

The UNFCCC (1992) and the Kyoto Protocol (1997)

The 1992 UNFCCC, adopted at the Rio Earth Summit, dropped the assessed contribution pool entirely. Instead, Article 4.8 of the Convention survived as a political compromise: it merely acknowledged the needs of vulnerable developing countries, listed insurance as an action that parties "shall give full consideration to," and provided no specific financial mechanism,

replacing the demand for restitution with vague promises of future cooperation. Traces of this proposal can still be found in article 4.8 of the 1991 Convention, which calls on parties to consider appropriate actions, including “including (...) funding, insurance and the transfer of technology” (United Nations, 1992).

The 1997 Kyoto Protocol established binding greenhouse gas emission reduction targets exclusively for developed Annex I parties, centring the regime on mitigation (UNFCCC, 1997). As a result of the 1992 UNFCCC and the subsequent 1997 Kyoto Protocol, L&D was effectively buried as a distinct policy pillar, as developed nations feared endless compensation claims; the international climate regime focused strictly on preventing future emissions and adapting to expected changes.

The Marrakesh Accords (2001) and the Adaptation Funds

To facilitate adaptation, negotiators established specific financial mechanisms, such as the Special Climate Change Fund (SCCF) managed by the Global Environment Facility (GEF), established in 2001, which financed the additional costs of interventions measured against a predetermined development baseline (UNFCCC, 2001). By channelling climate finance exclusively through adaptation, the United Nations Framework Convention on Climate Change effectively declared all climate harms as adaptable conditions.

This institutional design choice left irreversible L&D without a dedicated financial mechanism. Funding adaptation inherently fails to cover the residual losses that occur when climate impacts breach physical and social adaptation limits, forcing SIDS and LDCs to absorb the costs of L&D during this period.

The Cancún Adaptation Framework (COP16, 2010)

The 2010 Cancún Adaptation Framework formally included L&D in a COP decision text for the very first time (UNFCCC, 2010). While it also created the first Loss and Damage Work Programme, it placed it under the adaptation umbrella and treated it as a management problem.

It also explicitly avoids the issue and instead frames the issue as a need to "understand and reduce L&D", guiding the resulting work programme to focus on "risk management and reduction" alongside "risk sharing and transfer mechanisms such as insurance". Placing L&D inside the adaptation pillar locks the international response into a specific, restrictive path. If a treaty defines loss simply as an adaptation shortfall, it limits the available financial solutions to technical adjustments such as disaster risk reduction or micro-insurance.

Warsaw International Mechanism (COP19, 2013)

State of negotiations leading to WIM

Leading up to COP19, vulnerable states faced escalating disaster costs. Existing climate finance failed to cover permanent destruction. Developing countries needed a dedicated financial mechanism for irreversible climate impacts urgently, while developed nations feared unlimited financial exposure and suppressed treaty text establishing legal liability or mandatory compensation. This gridlock defined the political battleground.

Super Typhoon Haiyan proved this at COP19 as it struck the Philippines, causing massive destruction, providing concrete evidence that mitigation and adaptation fail to prevent extreme losses (Santos et al., 2015). The Philippine negotiator Yeb Sano's hunger strike during this event, appealing for support for the Philippines, created intense global pressure, and developing country delegates and civil society observers walked out in protest of industrialised nations' refusal to engage with the proposed financial mechanism, demonstrating that developing nations would no longer accept vague technical support in place of capital (Falzon et al., 2023; McGrath, 2013).

Developed nations, particularly the United States, the European Union, and the Umbrella Group, firmly opposed the establishment of a distinct financial mechanism due to domestic political constraints (Calliari et al., 2018, 2020). US negotiators operated under strict directives to block any text that could create legal liability or require compensation and explicitly rejected liability and compensation while supporting technical help (Calliari et al., 2018, 2020).

To neutralise demands, developed countries used specific tactics as documented by Calliari (2014, 2018b) and Falzon (2023). First, they tried to trap L&D within existing adaptation and disaster risk reduction frameworks, arguing that current institutions were sufficient, as seen in the Cancun Frameworks. Second, they used scientific uncertainty to dispute liability, arguing attribution problems prevented direct compensation. Finally, they attempted to shift focus from financial restitution to non-economic losses and risk management strategies. Their financial suggestions focused on weak alternatives such as private insurance schemes, thereby shifting the burden onto vulnerable populations.

These incompatible positions created severe instability. Developing countries demanded a new institution with financial power. Developed nations refused to concede on liability and fought

to restrict any new mechanism to information sharing and dialogue. This conflict forced an eleventh-hour deadlock, shaping the heavily compromised Warsaw International Mechanism.

Outcomes at WIM

Decision 2/CP.19 established the WIM with specific functions enhancing knowledge and understanding of comprehensive risk management approaches; dialogue, coordination, coherence, and synergies among relevant stakeholders; and action and support, including finance, technology, and capacity-building (UNFCCC, 2013, para. 5 (a), (b), (c)).

It grants the WIM authority to collect data and convene meetings but denies it the power to raise capital. While the third function mentions "finance", it mandates only the enhancement of support, not the provision of funds, establishing it strictly as a dialogue and knowledge mechanism, neutralising the demand for financial restitution.

Before Warsaw, the G77 and China submitted proposals demanding a dedicated mechanism comprising "means of rehabilitation and/or compensation" (G77 and China, 2013a, 2013b). The Alliance of Small Island States (AOSIS) demanded a financial facility with a compensatory component to address slow-onset events such as sea-level rise (AOSIS, 2012). Draft texts explicitly proposed a technical facility to redress unavoidable L&D alongside a financial facility to manage dedicated funding sources. Developed nations, led by the United States, meanwhile, actively deleted this vocabulary.

The final text of Decision 2/CP.19 gutted the proposed financial facility, leaving "only a work plan, without mention of a financial mechanism". "Compensation" and "liability" were entirely expunged from the final agreement. Negotiators inserted vague directives to "enhance understanding" and promote "comprehensive risk management". Developing countries demanded a standalone L&D mechanism independent of adaptation. The final text placed the WIM "under the Cancun Adaptation Framework".

This structural choice legally subordinated L&D to adaptation. It framed permanent destruction as a future management problem rather than a present harm requiring redress. To force consensus, negotiators deployed contradictory language. The preamble of Decision 2/CP.19 acknowledges that L&D "involves more than that which can be reduced by adaptation", but the text locates the mechanism within the adaptation framework, allowing developed nations to accept the text without accepting liability (UNFCCC, 2013).

The Paris Agreement

The 2015 Paris Agreement created a new global climate framework and aimed to limit global warming to well below 2°C above pre-industrial levels (UNFCCC, 2015b). It rests on three specific pillars: mitigation to cut greenhouse gas emissions, adaptation to manage future risks, and L&D to address unavoidable climate impacts

While Article 8 of the Paris Agreement is a victory, as it establishes L&D as a pillar of climate action, the text itself fails to secure mandatory finance. A close reading of Article 8 reveals that developed nations replaced binding obligations with aspirational statements. Because the treaty relies on voluntary cooperation rather than legal duty, the system cannot generate the necessary revenue to repair irreversible climate harm.

Article 8 opens by establishing boundaries that protect developed nations from legal exposure. In Article 8.1, parties merely "recognize the importance of averting, minimizing and addressing L&D" (UNFCCC, 2015b). The verb "recognize" acknowledges it but demands no specific action. The verb "recognize" acknowledges a reality but demands no specific action.

Article 8.2 states the Warsaw International Mechanism (WIM) "shall be subject to the authority and guidance" of the parties (UNFCCC, 2015b, 2016b). While this clause uses the binding modal verb "shall", it only regulates the administrative reporting structure of the WIM. It delegates no fiscal authority and does not force any party to fund the mechanism.

The core evasion of financial responsibility occurs in Article 8.3, which states that parties "should enhance understanding, action and support... as appropriate, on a cooperative and facilitative basis" (UNFCCC, 2015b, 2016b). In legal drafting, the choice of a modal verb determines the force of the law. Negotiators deliberately chose "should" rather than "shall". "Shall" creates a binding obligation. "Should" offers a recommendation. By choosing "should", the agreement reduces financial support to a discretionary political choice. Furthermore, the modifier "as appropriate" gives donor states an immediate exit, allowing them to withhold support whenever they deem it unsuitable. The phrase "cooperative and facilitative basis" reinforces this voluntary nature, legally severing the text from concepts of corrective justice or historical responsibility.

Article 8.4 defines the scope of this voluntary cooperation and states that areas of cooperation "may include" early warning systems, emergency preparedness, risk insurance facilities, and resilience building (UNFCCC, 2015b, 2016b). Again, the text uses the permissive modal "may"

instead of a directive command. This list limits expected responses to technical risk management and insurance schemes that transfer the financial burden back to vulnerable populations. It contains no operative provision that compels wealthy nations to grant debt-free funds for permanent losses.

To ensure Article 8 created no financial exposure, negotiators drafted a specific exclusion clause outside the treaty text. Paragraph 51 of **Decision 1/CP.21** explicitly dictates that Article 8 "does not involve or provide a basis for any liability or compensation". This clause removes the juridical link between causation and compensation (UNFCCC, 2016b).

By eliminating liability, the agreement prevents vulnerable states from demanding reparations for historical emissions by industrialised nations. Consequently, Article 8 remains formally isolated from Article 9, which governs binding climate finance for mitigation and adaptation.

The discussions behind the Paris Agreement

By the 2015 Paris talks, the G77 and China bloc offered a major concession and drafted Article 8, omitting "liability" and "compensation" (Vanhala & Hestbaek, 2016). However, the United States rejected this compromise as lead US negotiator Todd Stern stated the US refused the notion of liability (Vidal, 2015). US Secretary of State John Kerry stated the US's reluctance as follows: "We are not against [L&D]. We're in favour of framing it in a way that doesn't create a legal remedy because Congress will never buy into an agreement that has something like that...the impact of it would be to kill the deal" (Neslen & Harvey, 2016).

To appease the US, the French COP Presidency released a late-stage draft that inserted a liability ban directly into the treaty text, mandating that L&D proceed "in a manner that does not involve or provide a basis for liability or compensation" (Wewerinke-Singh & Salili, 2020). While Vanuatu recognised this trap and intervened during the final hours of the conference, Vanuatu's head of delegation called an emergency meeting with the Presidency. It threatened to block the consensus required to pass the Paris Agreement if the treaty retained the liability ban, halting the negotiations.

To break the deadlock, US delegates and small island leaders held closed-door meetings in which they abandoned the Presidency's draft treaty clause (Wewerinke-Singh & Salili, 2020). Instead, negotiators moved the liability exclusion out of the legally binding treaty and into the accompanying text as Paragraph 51 of Decision 1/CP.21.

Several small island states immediately exposed the political nature of this compromise. Upon signing the Paris Agreement, Vanuatu and others filed formal declarations stating their ratification did not constitute a renunciation of any rights under international law. By shifting the text to the decision document, developed nations secured their financial immunity, but developing nations kept the treaty text free of a permanent legal ban.

Developments after the Paris Agreement leading to 2022

In 2017, the WIM adopted its five-year rolling work plan but excluded new public finance, steering vulnerable countries toward private insurance and risk pooling (UNFCCC, 2017). High-emitting states replaced specific mandates with dialogues and commercial risk products, forcing poor nations to pay market premiums for their own protection while shielding wealthy nations from liability (Gewirtzman et al., 2018).

This approach failed on a scale comparable to that of actual disasters when Hurricane Maria struck Dominica in 2017, causing \$1.37 billion in damage. Yet, the regional insurance facility paid out only \$19.3 million (Richards & Schalatek, 2017).

Negotiators then offered technical advice rather than capital to delay binding commitments. At COP25 in 2019, parties established the Santiago Network to connect vulnerable countries with technical experts (UNFCCC, 2019). However, this body provided no new money to repair irreversible harm, focusing entirely on sharing knowledge and risk management strategies.

The refusal to provide concrete funding reached a breaking point at COP26 in 2021. The G77 and China bloc demanded the creation of a Glasgow Loss and Damage Facility to disperse funds directly (Falzon et al., 2023; McDonnell, 2023).

The United States, Australia, and the European Union rejected this proposal and offered the Glasgow Dialogue, a three-year discussion process with no mandate to secure actual funding. This tactic extended the period of talk without action, leaving vulnerable states without financial recourse as global temperatures continued to rise (Falzon et al., 2023; McDonnell, 2023).

Developments at COP27

With the failure of the Glasgow Dialogue coinciding with escalating climate catastrophes, such as the 2022 Pakistan floods, COP27 at Sharm el-Sheikh forced developing countries to accept

L&D finance onto the official agenda after overnight negotiations (Adger, 2023; Falzon et al., 2023; McDonnell, 2023).

Under intense pressure, developed nations conceded. Parties agreed to establish a dedicated Loss and Damage Fund. Although this decision ended decades of obstruction, the agreement leaves the exact funding sources unresolved, keeping the actual flow of money uncertain (Adger, 2023; Falzon et al., 2023; McDonnell, 2023).

At COP27 in Sharm el-Sheikh, parties agreed to establish new funding arrangements and a dedicated fund to assist particularly vulnerable developing countries affected by climate change, acknowledging an "urgent and immediate need for new, additional, predictable and adequate financial resources" to respond to both economic and non-economic harm (UNFCCC, 2023). However, the text serves only as an empty vessel, lacking operative provisions that define who pays, how much they pay, and the specific mechanisms that will generate revenue.

The COP27 decision deliberately does not mandate assessed contributions based on historical emissions or economic capacity, as developed nations refused to link the new fund to liability or compensation, making all financial transfers remain voluntary (Falzon et al., 2023; McDonnell, 2023; Siegele, 2023; UNFCCC, 2023). Furthermore, this framing of the text allows wealthy nations to classify existing humanitarian aid, loans, or private insurance schemes as L&D finance, thereby fragmenting the architecture and undermining the demand for a unified, grant-based mechanism.

To address these missing financial and structural elements, the decision delegates the design of the fund to a 24-member Transitional Committee, which will define the institutional arrangements, modalities, governance, and terms of reference (UNFCCC, 2023). Crucially, the committee must identify and expand on specific funding sources.

Developments at COP28

At COP28 in Dubai, parties adopted the Governing Instrument for the Fund for Responding to Loss and Damage (FRLD), marking a historic step toward providing financial assistance to developing nations highly vulnerable to climate change impacts (UNFCCC, 2024). The fund, operating under the UNFCCC and the Paris Agreement, is designed to address both economic and non-economic losses, including those from extreme weather and slow-onset events and is directed to fill priority gaps within the existing finance architecture (UNFCCC, 2024).

It establishes a 26-member Board to govern the money. It sets the rules for a new, independent secretariat, with the World Bank hosting the secretariat and serving as the interim trustee for four years (Board of the FRLD, 2024a; Board of the FRLD & IBRD, 2024). While developing nations wanted an independent institution, developed nations wanted World Bank control. To resolve this, the rules force the World Bank to grant direct access to developing countries and allow the Fund's rules to supersede Bank policies if they clash (Board of the FRLD, 2024a).

The summit also secured the first financial commitments. Donor countries pledged USD 792 million to the broader L&D funding arrangements, directing USD 661 million specifically to the FRLD (Board of the FRLD, 2024c). Germany, Italy, France, and the UAE led these initial pledges, each offering USD 100 million or more (Board of the FRLD, 2024c, 2024b). Additionally, Japan provided USD 10 million directly to the UNFCCC to cover the interim secretariat's setup costs (Board of the FRLD, 2024c). While these pledges allowed it to exist on paper, they accounted for less than 1 per cent of the hundreds of billions needed annually, leaving an unresolved gap between the Fund's goals and its resources (Heinrichs et al., 2025).

Developments at COP29

COP29 in Baku shifted the focus from creation to legal accountability as the conference approved formal arrangements between the COP, the CMA, and the FRLD Board (UNFCCC, 2025a). During this summit, the Board signed a Host Country Agreement with the Philippines, granting it independent legal personality and the capacity to negotiate contracts. By securing legal standing, the Board removed the final administrative barrier to hiring an Executive Director and forming the permanent secretariat.

The Baku summit also exposed the slow pace of revenue generation, with the COP issuing its first formal guidance to the Fund, urging donor countries to convert their pledges into signed contribution agreements quickly (UNFCCC, 2025a). Pledges grew marginally to approximately USD 820 million, aided by new commitments from Luxembourg, Spain, and the Walloon Region of Belgium (UNFCCC, 2025a).

To unite the fragmented L&D network, the FRLD co-convened the first annual high-level dialogue at the World Leaders Climate Action Summit, bringing together multilateral banks, United Nations agencies, and state leaders to align their tools (Board of the FRLD, 2024d).

Developments at COP30

At COP30 in Belém, the FRLD transitioned from planning to active funding, launching the first call for funding requests on the opening day of the summit (Board of the FRLD, 2026c, 2026b). This call operates under the Barbados Implementation Modalities (BIM). This short-term testing phase allocates USD 250 million across 2025 and 2026, distributing grants ranging from USD 5 million to USD 20 million to help vulnerable countries test local interventions and build national response systems (Board of the FRLD, 2025d, 2025a, 2026b). To protect the most exposed nations, the Board guaranteed that SIDS and LDCs would receive at least 50 per cent of this initial money.

The Belém summit also tested the Fund's ability to coordinate with external agencies, deploying a dedicated team to support the negotiations directly. To demonstrate that the Fund does not operate in isolation, the FRLD co-hosted an event with the Santiago Network and the Warsaw International Mechanism Executive Committee, showcasing how the three pillars of the L&D architecture (knowledge, technical assistance, and finance) combine to support developing countries (Board of the FRLD, 2026c). The call for funding requests remains open until 15 June 2026, marking the first time vulnerable states can access dedicated multilateral cash to rebuild after climate disasters (Board of the FRLD, 2026a).

Analysing FRLD in detail

The Governing Instrument gives the FRLD a broadly specific mandate, and while the mandate covers both economic and non-economic loss and damage, it explicitly targets impacts from extreme weather events and slow-onset events (UNFCCC, 2024). The text directs the Fund to support actions that complement immediate humanitarian relief, focusing instead on intermediate or long-term recovery, reconstruction, and rehabilitation (UNFCCC, 2024). The Fund also supports responses to climate-induced human mobility, including displacement and planned relocation (UNFCCC, 2024).

Paragraph 2 states that the Fund exists to assist developing countries that are particularly vulnerable to climate change in responding to economic and non-economic L&D, while Paragraphs 8 and 9 expand this scope to cover recovery, reconstruction, slow-onset events, and climate-induced displacement (UNFCCC, 2024). The Governing Instrument acknowledges that communities lose lives, cultural heritage, and territory, but doesn't provide a corresponding mechanism to compel historic polluters to pay for the losses.

To finance these operations, the Governing Instrument permits the Board to receive funds from a "wide variety of sources of funding, including grants and concessional loans from public, private and innovative sources. Critically, it sets no minimum capitalisation requirement (UNFCCC, 2024). The accompanying **decision 1/CP.28** urges developed countries to provide support, but explicitly frames these contributions as voluntary (UNFCCC, 2024). Contributors deposit payments in cash or promissory notes via individual contribution agreements, retaining complete discretion over funding volume and frequency. It also entirely omits mandatory financial tools, defining no formula for assessed contributions, setting no legally binding capitalisation target, and providing no mechanism for levies, thereby leaving the FRLD without an enforcement mechanism to penalise non-contributing parties.

The institutional design further limits the FRLD's autonomy, as **Decisions 1/CP.28 and 5/CMA.5** appoint the World Bank as the interim host and trustee for 4 years (UNFCCC, 2024). But while the Governing Instrument supersedes World Bank policies during disputes, the FRLD must bear the World Bank's direct and indirect cost-recovery fees, subjecting the Fund to the fiduciary architecture of an existing multilateral bank rather than establishing an independent financial entity with its own revenue-raising powers. (Board of the FRLD, 2025b, 2025b; Board of the FRLD & IBRD, 2024).

The agreement also locks in these constraints for years. Paragraph 55 mandates a periodic replenishment every four years, while paragraph 66 subjects the Fund to periodic reviews conducted by the COP and the CMA based on independent evaluations (UNFCCC, 2024). In addition, the COP will assess the World Bank's performance at the end of the four-year interim period before deciding whether to establish the Fund as an independent stand-alone institution and serve as the only formal mechanism for vulnerable states to contest the Fund's design, guaranteeing that developed nations face no immediate pressure to alter the voluntary model. (Board of the FRLD & IBRD, 2024; UNFCCC, 2024).

Years	Milestones	L&D Finance Advocacy	L&D Finance Obstructions
1991-2007	1991: Vanuatu, on behalf of AOSIS, proposed an international insurance for sea level rise. 1992: Adoption of the UNFCCC. Insurance (by AOSIS) is listed under Article 4.8 only as a meeting option. 2001: Marrakesh Accords established LDCF and SCCF.	Vanuatu and AOSIS pushed for L&D inclusion and proposed a collective-loss-sharing insurance scheme to compensate victims of sea-level rise. They framed L&D as adverse effects and risk transfer.	Developed nations excluded the specific insurance proposal from final agreements and the 1992 Convention. No funding mechanism was established under Article 4.8. The article conflated L&D with the impacts of response measures for oil producers.

	2007: Bali Action Plan Decision 1/CP.13 formally included L&D terminology in negotiation text.	They argued for funding based on historical responsibility, emissions and capacity GNP. They pushed for new, additional, and adequate revenue.	Global North shifted discussions to informal workshops and technical programs to avoid financial commitments.
2008-2013	2008: AOSIS submitted a multi-window mechanism proposal including insurance, rehabilitation and compensation, and risk management. 2010: Cancun Agreements established a work program on L&D under the Cancun Adaptation Framework. 2012: Doha COP18 Decision 3/CP.18 moved toward institutional arrangements. 2013: Warsaw International Mechanism (WIM) launched in COP19.	AOSIS, LDCs, and the African Group coalition, later unified as the G77 plus China, proposed compensation and rehabilitation mechanisms. They demanded strong negotiation texts in Warsaw on a dedicated institutional body and financial assistance to address unavoidable impacts. A Philippine negotiator's 2013 hunger strike drew global media attention to the demand for a solidarity fund.	Developed nations, led by the US and UK, weakened the WIM language to ensure it remained dialogue-based and restricted to knowledge sharing. They forced the WIM to be established under the Cancun Adaptation Framework to restrict its scope and avoid separate finance tracks. The UK cut specific funding support for L&D-related initiatives. They conflated L&D with adaptation to deflect from historical responsibility.
2014-2015	2014: WIM's initial two-year work plan was adopted. 2015: Paris Agreement Article 8 established L&D as a standalone concept or pillar. 2015: Decision 1/CP.21 Paragraphs 51 and 52 accompanied the Paris Agreement.	Developing nations secured recognition of L&D as a third pillar distinct from Article 7 adaptation. AOSIS linked L&D to ambitious temperature goals. They broadened the framing to include non-economic losses such as life, culture, and identity, as well as migration and displacement. NGO and media pressure increased around climate justice.	The US and Australia resisted until the final hours and blocked linkage between Article 8 and mandatory finance. Negotiators explicitly excluded liability or compensation from Article 8. Efforts were made to steer L&D finance toward private insurance and pilot risk transfer schemes rather than public grants.
2016-2022	2016: First WIM review held in Marrakesh. 2017: WIM five-year rolling workplan and Fiji Clearing House for Risk Transfer established. 2019: Santiago Network for L&D established at COP25. 2021: The Glasgow Dialogue was established after developed nations blocked a finance facility at COP26. 2022: COP27 Decision 2/CP.27 established the FRLD	G77 plus China unified to demand a dedicated L&D Finance Facility and for L&D to be a standalone agenda item. They secured inclusion of L&D in the Global Stocktake. They pushed for new and additional financing beyond ODA. The Scottish Government made the first voluntary commitment in 2021. The V20 group advocated for regional risk pools and InsuResilience.	Developed nations, including the US, EU, and Australia, initially blocked the finance facility proposal in Glasgow and instead offered the non-binding Glasgow Dialogue. They promoted insurance-based alternatives such as Global Shield and InsuResilience. They delayed funding decisions through technical work programs and slowed operationalisation of the Santiago Network. They pushed for a broad donor base to include emerging economies to dilute historical responsibility.
2023-present	2023: Finalised fund design, and FRLD was operationalised at COP28. 2024: Board Meetings B1 to B4 held. Ibrahima Cheikh Diong	G77 plus China secured institutionalisation of the fund and pushed for grant-based, predictable, new, and additional finance.	The Global North insisted on the World Bank as the interim host and trustee despite concerns about autonomy and bureaucracy.

appointed Executive Director. The Philippines was selected as the host country.	They pushed for direct access for subnational and national entities, direct budget support, and community-led funding.	They preferred voluntary contributions over assessed obligations based on historical emissions.
2024: Pledges mobilised for FRLD, and signed with the World Bank.	SIDS and LDCs secured a 50 per cent allocation floor in the Barbados Implementation Modalities.	They continued to resist liability and compensation, as well as human rights-based legal framing.
2025: First Annual High-Level Dialogue scheduled and BIM launched.	Advocacy focused on rapid delivery and country-led interventions.	Initial pledges remained limited relative to projected needs in the hundreds of billions of dollars.
2026: Expected first funding round and initial project approvals.		

Table 4.1: Five Periods of Contention over Loss and Damage Finance in the UNFCCC and Positions Taken by Advocates and Opponents, inspired by Falzon et al. (2023).

Analysing the Discourse around L&D as a whole

Despite the Vanuatu proposal, the 1992 Convention and the later WIM deliberately excluded compensation and liability for L&D, framing it instead as a subordinate risk-management problem focused on knowledge and dialogue. As demonstrated earlier, the 2015 Paris Agreement cemented this legal immunity for mandatory finance through Article 8 and accompanying decisions at COP21, defaulting to a cooperation model grounded in distributive justice. When wealthy nations contribute out of solidarity rather than legal obligation, they dictate the terms, timing, and scale of their payments. Vulnerable states must rely on unpredictable funding cycles tied to donor nations' political will.

Developed countries promote private insurance to fill the public finance gap. Commercial insurance based on mutuality compels communities at risk to pay premiums, thereby shifting the financial burden back onto vulnerable populations and contravening common but differentiated responsibilities. Insurance proves ineffective for slow-onset events like sea-level rise because they constitute certainties, not risks.

The reliance on market mechanisms systematically ignores elements that it cannot adequately price. NELD encompasses the loss of life, health, territory, culture, indigenous knowledge, and biodiversity. Insurance and compensation models deliberately exclude these profound losses due to the strict requirement for monetary metrics, rendering NELD a mere residual category. By treating NELD as intangible, the current governance framework consequently creates an **ontology of ignorance**, sidelining human consequences, obscuring fundamental root causes and marginalising legitimate demands for restorative justice.

The UNFCCC's deeply ingrained history directly determines the intrinsic limitations of the new FRLD. Decision 1/CP.21 effectively removed all notions of liability, leaving the FRLD

significantly lacking the statutory authority to mandate substantial contributions from high-emitting states legally.

The existing architecture deliberately relies on varied funding arrangements, which unfortunately disperse responsibility across a wide range of entities. This fragmented, solidarity-based approach, therefore, fundamentally demands voluntary pledges rather than legally assessed contributions. The deliberate exclusion of liability unequivocally ensures that the FRLD will remain discretionary aid until innovative revenue instruments that do not rely on voluntary cooperation are meticulously developed.

Analysis of Negotiation and Interest Blocs

The consensus process of global climate governance forces vulnerable states to trade legal rights for institutional survival. While developing nations seek corrective justice for permanent climate destruction, wealthy nations refuse to pay. To prevent negotiations from collapsing, vulnerable states consistently abandon their demands for mandatory compensation. This dynamic produces empty institutions that lack fiscal authority.

The matrix reveals a clear trajectory of "demand migration" among the developing nations, as the AOSIS and the LDC Group moved from demanding strict liability to accepting frameworks built entirely on voluntary cooperation. Vanuatu, on behalf of AOSIS, initially demanded a mandatory, emissions-based insurance mechanism funded by wealthy nations, consistently framing it as corrective justice until 2012 (Calliari, 2018b; Intergovernmental Negotiating Committee, 1991). However, by 2023, these same blocs accepted the FRLD, moving from a demand for mandatory, liability-based funding to an institution that relies exclusively on voluntary pledges and explicitly rejects liability (Falzon et al., 2023).

Yet, this migration does not mean their normative beliefs changed. Throughout the 33 years, submissions from AOSIS and the LDCs continue to invoke historical responsibility and climate justice. Instead, their strategic calculus shifted (Wewerinke-Singh, 2023). Negotiators from vulnerable states calculated that securing a formal mechanism with weak fiscal authority was better than securing no mechanism at all (Calliari, 2018b; Wewerinke-Singh & Salili, 2020).

The primary inflexion point occurred at COP19 in 2013, when AOSIS and the LDC Group entered the Warsaw negotiations, demanding a mechanism with a clear mandate for compensation (Calliari, 2018b). Developed nations, particularly the United States, outright

rejected this, leading developing nations to accept the WIM, a body stripped of authority and restricted to sharing knowledge (Calliari, 2018b).

Before 2013, the European Union and the Umbrella Group used hard opposition, rejecting L&D as a distinct category and forcing all discussions into the adaptation framework to avoid any implication of liability (Calliari, 2018b). After COP19, their strategy evolved into "soft opposition, as they now accept the L&D label but actively strip the resulting mechanisms of any financial obligation (Falzon et al., 2023). By replacing clear legal duties with ambiguous text regarding "support" and "dialogue," they appear cooperative while avoiding payments.

At COP26 in 2021, the G77 and China demanded a Glasgow Loss and Damage Facility to disburse funds directly, but this was blocked, with the alternative being the "Glasgow Dialogue" (Falzon et al., 2023). By COP27, escalating physical disasters, such as the 2022 Pakistan floods, combined with intense pressure from the G77, forced the EU and Umbrella Group to shift tactics and agree to establish a dedicated fund. However, they immediately applied their soft opposition strategy, ensuring the COP27 decision explicitly separated the new fund from liability or compensation, allowing the FRLD to exist but functioning as a mechanism for voluntary charity rather than a tool for corrective justice (Falzon et al., 2023).

Evolution of Instruments Used

Instrument	Mechanism	Status of Application
Grants	Direct cash transfers require no repayment.	Status: Adopted. Designated as the primary instrument for the start-up phase of the new FRLD (Board of the FRLD, 2025c). Developing nations demand grants to avoid sovereign debt.
Concessional Loans	Debt issued at below-market interest rates.	Status: Adopted Included in the FRLD Governing Instrument for future use (Board of the FRLD, 2025c). Vulnerable states oppose loans because they force victims to borrow money to recover from disasters, they did not cause, driving them into debt crises.
Insurance & Risk Pooling	Policyholders pay premiums to transfer risk to private insurers or regional pools.	Status: Adopted and heavily promoted by developed nations. Fails for inevitable slow-onset events and cannot value NELD
Catastrophe Bonds	High-yield debt instruments that forgive principal if a specific disaster trigger is met.	Status: Adopted in limited cases. Provides rapid liquidity but incurs high fixed costs and covers only sudden, unpredictable catastrophes, ignoring slow-onset hazards.
Contingency / Pre-arranged Finance	Pre-approved credit lines that governments draw from immediately after a disaster.	Status: Adopted. Offers rapid post-disaster liquidity but ultimately adds to sovereign debt burdens.

Direct Budget Support	Financial transfers are sent directly to national treasuries to fund public policy or emergency response.	Status: Adopted. Listed in the FRLD Governing Instrument. Allows rapid, country-led disbursement following extreme weather events.
Levies and Taxes	Mandatory surcharges on international airline travel, bunker fuels, financial transactions, or fossil fuel extraction.	Status: Proposed but stalled. Offers predictable, new revenue aligned with the polluter-pays principle. High-emitting nations consistently block implementation.
Solidarity Funds	Pools of money are built from voluntary pledges	Status: Adopted. Operates outside strict liability. Relies entirely on discretionary charity, which makes revenue unpredictable and usually inadequate.
Compensation Facilities	Mandatory payments are extracted from nations based on their emissions and wealth.	Status: Rejected. Developed nations strictly veto any mechanism that admits legal liability or mandates restitution.
Equity Investments	Capital is provided in exchange for ownership shares and financial returns.	Status: Authorised in FRLD Submerged land and destroyed infrastructure generate no dividends, leaving equity investors with no underlying asset to hold.

Table 4.2: Every major financial mechanism discussed within the United Nations Framework Convention on Climate Change (UNFCCC).

Evaluating only adopted climate agreements hides how negotiators dismantle binding financial mechanisms. To understand the actual trajectory of L&D governance, one must examine the proposals that failed. When analysing the rejected instruments, the historical pattern of climate finance negotiations reveals a consistent strategy by wealthy nations to reject mandatory obligations, leading to a continuous weakening of ambition in successive proposals.

From vetoing AOSIS's 1991 demand for a fund based on assessed contributions to removing mandatory revenue mechanisms from later demands, developed nations have consistently favoured voluntary rather than binding duties.

This deliberate rejection has resulted in architecturally weaker instruments, shifting from a structured, compensatory pool to mechanisms with no financial component, and ultimately to the FRLD in 2022-23, which relies entirely on voluntary pledges.

Furthermore, a "substitution strategy" is employed, in which demands for financial mechanisms are deflected and replaced with non-financial alternatives such as knowledge-sharing committees, procedural delays, or technical assistance, effectively substituting capital demands with research mandates and advice.

Year	Proposing party	Instrument type	Obligation	Outcome	Grounds for rejection or dilution
1991	AOSIS	International insurance pool with assessed contributions.	<i>Mandatory</i>	Rejected	Liability concerns led to being dropped entirely from the UNFCCC in 1992. Replaced by Article 4.8 recognition with no financial mechanism.
2001	AOSIS / G77	Compensation and rehabilitation fund for irreversible impacts	<i>Mandatory</i>	Rejected	Subsumed into the SCCF as an adaptation instrument, the compensation element was removed.
2008	AOSIS	insurance + compensation + risk management + rehabilitation	<i>Mandatory</i>	Rejected	Donor opposition; WIM substituted, dialogue and knowledge body only, no financial authority.
2013	G77 and China	Dedicated financial facility with a compensatory component for unavoidable and slow-onset impacts	<i>Mandatory</i>	Rejected	Final Decision 2/CP.19 stripped the financial mechanism entirely. "Compensation" and "liability" expunged. WIM is confined to knowledge sharing.
2021	G77 and China	A dedicated fund to disburse grants to affected countries	<i>Mandatory</i>	Rejected	Replaced with non-binding Glasgow Dialogue, a three-year discussion process with no funding mandate.
2023	AOSIS / LDC Group	Independent FRLD with mandatory capitalisation targets, assessed contributions, and levy mechanisms.	<i>Mandatory</i>	Partially rejected	World Bank hosting adopted over independence; voluntary contributions only; no levy mechanism; no capitalisation floor. The Governing Instrument excludes all mandatory revenue authorities.

Table 4.3: Every major financial instrument proposed during L&D negotiations that final decision texts either rejected or neutralised. A detailed table of all instruments will be included in the Annexures.

Current Financial Status of FRLD

Category	USD. (millions)
Cash actually received	456.75
Contributions signed but unpaid	144.79
Pledges (expressions of intent only)	226.92
Total contributions (as of December 31, 2025)	828.47
Total funds ready for immediate usage	420.79

Table 4.4: Summary of the Financial Status of FRLD

The December 2025 Trustee Report (2025) provides the clearest possible window into the gap between the fund's purpose and its operational reality. Total pledges and contributions to the FRLD as of December 31, 2025. stand at USD. 828 million, of which USD 456 million has been received in cash (World Bank, 2025) but is insufficient as conservative academic estimates place annual L&D costs for developing nations at \$400 billion by 2030, rising steeply

thereafter (Markandya & González-Eguino, 2019). The FRLD's total potential resources represent approximately 0.1% of a single year's projected need.

Nearly half the headline number (\$371.71 million) consists of resources not yet received, a substantial portion of which are pledges that carry no binding legal obligation. The substitution strategy documented in rejected proposals from 1991 onwards has, in this iteration, entered the capitalisation architecture itself: headline numbers are announced to signal ambition, while the underlying legal structure permits non-delivery.

Contributor-Level Analysis

Italy pledged EUR 100 million (USD. 117.47 million). The European Commission pledged EUR 25 million (USD. 29.37 million), none received as of December 31, 2025 while the UAE, which made its pledge during COP28 on its home territory, has transferred only USD 15 million of a USD 100 million pledge, leaving USD 85 million outstanding (World Bank, 2025).

The United States, historically the primary blocker of mandatory L&D finance (Calliari, 2018b; Falzon et al., 2023; McDonnell, 2023), contributed USD 17.56 million, reflecting both its constrained political willingness and the absence of any assessed contribution mechanism that would have required a proportionate share based on historical emissions or GDP. Under the 1991 AOSIS formula (50% by GNP, 50% by CO₂ share), the US obligation would have been orders of magnitude higher. With the FRLD Board deciding to set aside USD 250 million for the Barbados Implementation Modalities call for funding requests, pending subsequent funding decisions, there is an urgent requirement for more funds to be paid out for the FRLD.

The World Bank Hosting Problem, Expressed in Data

The FRLD's assets are managed not by an independent climate body but as part of the World Bank's commingled investment portfolio. The undisbursed cash balance is maintained in a commingled investment portfolio for all trust funds managed by IBRD, with funds managed such that the expected maximum loss in the worst 1% of cases will not exceed 0.25% at the portfolio's investment horizon (World Bank, 2025).

From inception through December 31, 2025, the FRLD Trust Fund earned investment income of approximately USD 16.28 million, reflecting a portfolio return of 4.96% (World Bank, 2025). It capitalised on the principle that polluters owe a debt to climate victims, generating returns by investing in the same financial system that finances the fossil fuel economy. Whether

those holdings include fossil fuel bonds or not, the fund's undisbursed balance generates returns while the communities it serves continue to experience irreversible losses.

The UK's contribution (GBP 40 million delivered as a promissory note rather than cash) creates a USD 4.04 million currency risk reserve that directly reduces funds available for FRLD Board decisions (World Bank, 2025). The World Bank's risk management conventions, designed for a broad portfolio of development lending, draw on project-eligible resources from a fund intended for victims of climate disasters.

DISCUSSION

The evidence in the findings produces one answer to the primary research question. The normative shift in UNFCCC texts from liability to cooperation structurally limits the FRLD's authority to enforce mandatory revenue measures. The distinction between a structural constraint and a political failure is not semantic, as political failure is reversible. If the FRLD's inadequacy came from weak leadership, poor advocacy, or an unfavourable political climate in major donor countries, then better negotiators or a change of government could fix it. Structural constraints work differently. They are enshrined in the institution's founding documents and do not yield to political goodwill. Changing them requires amending the architecture, which means reopening the agreed text.

The FRLD Governing Instrument confirms that this is a structural problem. Paragraph 2 authorises the Fund to assist particularly vulnerable developing countries, but it sets no minimum capitalisation requirement and defines no formula for assessed contributions (UNFCCC, 2024). Decision 1/CP.28 explicitly frames donor contributions as voluntary (UNFCCC, 2024). The Fund receives money through individual contribution agreements that give donors full discretion over volume and timing. The FRLD cannot require mandatory contributions not because its leadership has failed to ask for them, but because its founding document does not authorise it to do so.

The fiscal consequence is measurable. Conservative projections place annual loss and damage costs for developing nations at USD 290-580 billion by 2030, rising to USD 1.1-1.7 trillion by 2050 (Markandya & González-Eguino, 2019). Total pledges to the FRLD as of December 2025 stood at USD 822 million, of which USD 413 million had been received in cash (World Bank, 2025). The Fund's current capital is less than 0.2% of the projected need, a structural gap that will persist due to its lack of mandatory revenue authority.

Incremental improvements to the FRLD's governance, transparency, or operational efficiency will not resolve the funding gap. Only two pathways can: redesigning the mechanism to include some form of mandatory obligation, or building parallel mechanisms outside the FRLD on legal foundations (Wewerinke-Singh and Salili, 2020).

Why Developed Countries Accepted Cooperative Measures

When analysing why did vulnerable states abandon thirty years of advocacy for climate justice, five specific political and economic factors arise.

Bargaining Asymmetry

The UNFCCC requires consensus. This structure grants powerful nations veto power over any text mentioning compensation or liability. While vulnerable groups like the AOSIS exercise moral leadership, they lack material bargaining power, as SIDS and LDCs send small delegations that cannot cover negotiating tracks or outlast wealthy nations in closed-door sessions (Falzon et al., 2023). Powerful states also impose strict red lines, as shown by how the US rejected liability in earlier analysis (Calliari et al., 2018). Asymmetric dependence forces dependent nations to accept the minimum terms offered to secure any agreement.

Geopolitical Fatigue

Developed countries spent thirty years deploying obstructionist tactics and used soft bargaining strategies to trap L&D discussions inside adaptation frameworks and pushed endless technical dialogues to delay action (Falzon et al., 2023). This delay created profound geopolitical fatigue, as vulnerable nations realised that holding out for a liability mechanism meant enduring perpetual workshops while receiving zero capital.

Funding Urgency

When catastrophic shocks destroy national infrastructure, governments require instant liquidity to survive and fund disaster relief (McNamara & Jackson, 2019). They cannot wait decades for international courts to assign historical responsibility (Otto et al., 2017). The urgent need for post-disaster capital overrides the pursuit of ideal justice. Developing nations accepted voluntary, grant-based mechanisms because inadequate funding today keeps citizens alive, whereas theoretical legal victories do not.

Realism versus Idealism

The ideal of corrective justice remains the baseline for developing nations, as continuous submissions demanding historical responsibility show (Calliari, 2018). Negotiators calculated that establishing a formal mechanism provided a stronger foundation than having no mechanism. By accepting "constructive ambiguity," they secured the Warsaw International Mechanism and Article 8 of the Paris Agreement (Calliari, 2018). In Paris, nations dropped

their opposition to the liability exclusion—codified in Paragraph 51 of Decision 1/CP.21—because rejecting the treaty meant losing Article 8 entirely (Falzon et al., 2023).

Strategic Incrementalism

Developing countries used demand migration as a deliberate strategy. They traded legal rights for institutional survival. Dropping the word "liability" was the exact price of admission to get loss and damage recognised as a distinct pillar of climate action under the UNFCCC (Roberts & Huq, 2015). They agreed to cooperative language so the Fund for Responding to Loss and Damage (FRLD) could launch. This secured an institutional foothold they plan to reform from the inside over time.

Constructive Ambiguity as Destructive Ambiguity

This concept of destructive ambiguity extends from Vanhala's (2023) constructive ambiguity analysis. Vanhala's governance study identified fiscal consequences as a concern, but did not develop the mechanism in institutional economics terms. This concept corrects the usual, often praised, scholarly view of constructive ambiguity, as discussed by Vanhala (2023) and Calliari et al. (2018), who treat ambiguity as a tool for agreement in multilateral negotiations. While it does help in arriving at a consensus, destructive ambiguity does not allow examining its long-term effects on authority and finance.

This underfunding arises because the more demanding standard, namely liability or obligation, requiring explicit and affirmative legal language to operate. The voluntary standard, based on risk management and cooperation, is established simply through the absence of such binding provisions. This asymmetry creates a systematic bias within treaty design, where ambiguous language functions as a lacuna that favours the party seeking to reduce its obligations, transforming a short-term negotiating tool into a long term structural fiscal constraint.

The FRLD's mandate, to address annual losses projected in the hundreds of billions, is therefore systematically undermined by the operative language of its own Governing Instrument. This study defines this outcome as destructive ambiguity, where ambiguity enables agreement while simultaneously constraining the institution's capacity to mobilise adequate and predictable finance. This in financial contexts, does not facilitate development but instead establishes the lower cost institutional pathway benefitting only few parties.

The Free-Rider Problem

Major historical emitters that have not pledged contributions face no enforcement consequences, as no compliance or penalty exists within the current framework (Falzon et al., 2023). Åberg and Jeffs (2022) show that key contributors to global climate finance frequently make large pledges but disburse only a portion of committed funds. Van Schie et al. (2024) demonstrate that such gaps are not incidental but are structurally embedded within voluntary financing systems.

The 1991 AOSIS proposal provides a clear burden sharing formula, allocating 50 per cent based on gross national product and 50 per cent based on CO₂ emissions (Intergovernmental Negotiating Committee, 1991). Under such a framework, the United States would bear the largest financial responsibility due to its combined economic size and cumulative emissions. However, by December 2025, the United States had pledged only USD 17.56 million to the FRLD, reflecting both limited political commitment and the absence of any binding contribution mechanism (World Bank, 2025). The behaviour of other major donors reinforces this pattern, with Italy pledging EUR 100 million without disbursement by year end 2025, and the European Commission pledging EUR 25 million with no recorded disbursement.

Limits of Political Realism

Defenders of the cooperation model cite attribution uncertainty as its scientific justification. Establishing a direct causal link between a specific country's emissions and a local weather event remains technically challenging, and reliable long-term observational data is scarce precisely in the Global South, where the most vulnerable countries exist (Huggel et al., 2016). Because the legal standard for liability traditionally requires such a direct causal link, attribution uncertainty is said to render compensation frameworks legally impossible, and a cooperative model is presented as the scientifically honest alternative (Simlinger and Mayer, 2018). This does not hold. Attribution science now confidently measures how human activity increases the probability and intensity of specific weather events, and legal frameworks do not require absolute certainty to assign responsibility. Courts handle diffuse causation through a 'modified general causation test,' which requires only that greenhouse gases generally cause damages and that a causal link is probable (Simlinger and Mayer, 2018). Imperfect event-specific attribution does not preclude liability. Scientific uncertainty here functions as a political shield, not a scientific constraint.

The political feasibility case is more candid. Developed countries categorically reject any framework implying legal liability, and during the Paris Agreement negotiations the United States warned that any reference to compensation would 'kill the deal' (Calliari et al., 2018). A strict liability framework, the argument runs, would have collapsed negotiations and left vulnerable states with nothing. The cooperation model, codified through the 'constructive ambiguity' of Article 8, kept developed nations at the table (Vanhala, 2023). Yet the consensus it secured is largely empty. Excluding liability eliminates the legal mechanism required to mandate financial contributions, and the Paris Agreement's explicit rejection of liability ensures that funding remains voluntary and unpredictable (Simlinger and Mayer, 2018). Treating L&D as a cooperative risk-management issue strips the FRLD of its fiscal authority. What reads as constructive ambiguity at the negotiating table produces destructive ambiguity in the financial mechanism (M2022BSASS020, 2022).

A third line of defence frames the cooperation model as a solidarity approach that encourages broad participation without hostile legal battles. L&D support is presented as a humanitarian response to undeserved harm rather than a legal penalty, and private sector instruments such as catastrophe bonds and regional insurance pools like the Caribbean Catastrophe Risk Insurance Facility are offered as evidence that diverse finance can be mobilised without state obligation (Linnerooth-Bayer et al., 2019, Wallimann-Helmer, 2023). In practice, solidarity without obligation invites free-riding and shifts the financial burden onto the victims (Åberg and Jeffs, 2022). Developed nations routinely re-label existing official development assistance as L&D finance, erasing additionality (Weikmans and Roberts, 2019). Market-based insurance compounds this by requiring vulnerable countries to pay premiums for risks they did not create, violating the principle of common but differentiated responsibilities (Nordlander et al., 2019). Solidarity models convert restitution into discretionary charity.

Pathways Forward

Under current institutional constraints, solidarity levies on international aviation and maritime transport present the most viable revenue pathway and are linked to economic activity rather than state responsibility, allowing them to circumvent the liabilities in Decision 1 CP.21. They offer to generate predictable finance without requiring formal attribution of legal responsibility (Pill, 2022). Previous proposals have stalled due to political resistance rather than legal infeasibility, suggesting that reframing them under a solidarity-based narrative may improve their acceptability compared to explicitly liability-based approaches.

Multilateral Development Bank channelling provides only a partial and temporary solution. While MDB resources could expand short term funding availability, their reliance on loan-based instruments presents a fundamental limitation (CPI, 2023). As Adeniyi (2025) argues, financing L&D through debt instruments imposed on vulnerable countries raises serious equity concerns and risks deteriorating sovereign debt vulnerabilities. The distinction between MDB finance and dedicated L&D funding is therefore substantive rather than merely quantitative.

Private and philanthropic finance, on the other hand, while permitted under the Governing Instrument's reference to a wide range of sources (UNFCCC, 2024), cannot operate at the required scale. Existing global philanthropic flows for climate adaptation remain limited, and the absence of revenue generating returns further constrains private sector participation.

The scheduled review of the World Bank's hosting arrangement provides a critical institutional opportunity. Amendments to the Governing Instrument could introduce provisions for solidarity-based revenue streams, periodic funding adequacy reviews, or alternative hosting arrangements. These changes require COP level decisions rather than amendments to the Paris Agreement itself (Schalatek, 2025). The Country Coalition Position Matrix indicates a shift among developed countries from explicit opposition to more indirect forms of resistance, where L&D is accepted in principle but deprived of fiscal authority in practice.

Whether increasing climate impacts and operational pressures on the Fund will shift this position towards support for mandatory contributions remains an open political question. This study does not resolve that question but reframes it by identifying the legal and institutional constraints that must be addressed.

Situating the Findings in the Literature

The findings confirm Calliari's (2018) discourse analysis of party positions in pre-Paris negotiations. The liability/cooperation divide she documented in that period has been reproduced in the FRLD's institutional architecture exactly as her analysis predicted. This study extends her account longitudinally: the same divide operates across the full period from 1991 to 2024, and it produces measurable fiscal consequences at the level of the Governing Instrument.

The findings confirm Gewirtzman et al.'s (2018) argument that conventional instruments are inadequate for L&D. This study extends that argument by showing that the instruments that would be adequate have been systematically rejected over 33 years, not merely overlooked.

The rejected instruments register provides the first comprehensive record of this pattern across the full history of negotiations.

On institutional design, this study fills the gap identified in the existing literature, which characterised the liability exclusion in Decision 1/CP.21 as politically significant but did not trace its translation into the specific clauses of the FRLD Governing Instrument. The clause-level fiscal authority assessment does this directly, establishing a documented chain from the 2015 negotiating outcome to the 2023 institutional design.

CONCLUSION

This thesis tracks the evolution of Loss and Damage governance under the UNFCCC from 1991 to 2024, showing how competing interpretations of responsibility have shaped institutional design and financial outcomes over time. Rather than treating L&D as a purely normative or political issue, the analysis demonstrates that legal language and institutional structure directly determine the capacity of global mechanisms to mobilise finance at scale. The findings establish that textual choices are not secondary features of negotiation, but central determinants of fiscal authority and institutional effectiveness.

The original 1991 AOSIS proposal set out a clear liability-based framework with mandatory payments linked to emissions and economic capacity. Subsequent phases steadily moved away from this position. The 2010 Cancún Framework incorporated L&D within adaptation, reducing its distinct legal standing. The 2013 Warsaw International Mechanism limited the institutional mandate to knowledge generation and coordination. The 2023 FRLD Governing Instrument explicitly excluded liability in its founding text (UNFCCC, 2024). Analysis across the corpus shows a decline in liability-linked terms such as compensation, restitution, polluter pays, and state responsibility, alongside a parallel increase in cooperation-based language. This shift is consistent across negotiating texts, decisions, and institutional instruments.

The Country-Coalition Position Analysis reveals a stable divide in negotiating positions. Developing country blocs, including AOSIS and the LDC Group, initially demanded mandatory, emissions-based payments but gradually accepted an institutional outcome based on voluntary contributions (Falzon et al., 2023). This change does not indicate a shift in normative position, as references to historical responsibility and climate justice remain consistent in their submissions (Calliari, 2018). Instead, it shows an adjustment to political constraints, where a formally established but financially limited institution was accepted as preferable as opposed to continued deadlock. Developed country blocs show a parallel shift from hard opposition of L&D to conditional acceptance, while consistently limiting its financial authority at each stage of negotiation.

The Instrument Evolution Table provides further evidence of this pattern by documenting the systematic exclusion of mechanisms capable of generating predictable and adequate finance. Across all phases, proposals that included mandatory contributions or automatic revenue generation were either rejected or weakened. From the 1991 proposal for assessed contributions

to later proposals for dedicated financial facilities, each iteration reflects a reduction in fiscal ambition. This pattern demonstrates that the limitation of financial capacity is not incidental but results from repeated and consistent design choices over time.

The clause-level assessment of the FRLD Governing Instrument confirms the institutional consequences of these choices. The Fund has no authority to set mandatory contribution levels, impose levies, or link financial obligations to emissions or economic indicators. The liability exclusion in Decision 1 CP.21 is directly reflected in the absence of fiscal authority within the Governing Instrument. The framework does not establish a funding floor, define contribution rules, or provide enforcement mechanisms for non-compliance (UNFCCC, 2024). As a result, the Fund relies entirely on voluntary pledges and a broad set of non-binding funding sources.

The normative shift from liability to cooperation therefore produces a structural constraint on the Fund's capacity to generate adequate and predictable finance. This is not simply a political compromise but a design outcome embedded within the legal architecture of the regime. By removing binding language and enforceable mechanisms, the institutional framework limits itself to forms of finance that are inherently unstable and insufficient relative to the scale of projected losses. The funding gap is thus not a temporary shortfall but a predictable outcome of the governing design.

This study makes three main contributions to the literature. First, it extends existing discourse-based analyses by providing a longitudinal account of L&D negotiations across the full period from 1991 to 2024, moving the field from qualitative observation to empirically grounded analysis. The integration of textual analysis with position mapping and instrument tracking allows for a systematic examination of how negotiating positions translate into institutional outcomes. This approach demonstrates that discourse is not only reflective of political positions but also constitutive of institutional design.

Second, the study introduces the concept of destructive ambiguity as a refinement of existing interpretations of constructive ambiguity in multilateral negotiations. When the subject of contention is fiscal authority, ambiguity does not simply enable agreement but constrains the range of institutional outcomes that remain available. In this context, ambiguous language functions as a mechanism that favours lower-cost, voluntary arrangements while preventing the adoption of binding financial commitments. This finding challenges the prevailing assumption that ambiguity facilitates gradual norm development, showing instead that it can entrench structurally weaker institutional configurations over time.

Third, the study contributes to institutional design and public finance literature by demonstrating that financial capacity is directly shaped by textual provisions within governing instruments. An institution that relies on voluntary pledges will generate insufficient and unpredictable revenue if its founding document confers no authority to mandate contributions. This shifts the analytical focus from implementation failure to design limitation, reframing underfunding as a structural outcome rather than a consequence of weak political will or administrative inefficiency.

The four-year review of the World Bank's hosting arrangement therefore represents a critical institutional moment rather than a routine administrative process. It provides an opportunity to reassess the design of the Fund and to introduce mechanisms capable of generating more predictable revenue streams. Solidarity-based levies on international aviation and maritime transport present a viable option, as they are linked to economic activity rather than state responsibility and therefore do not contravene the liability exclusion in paragraph 51 of Decision 1 CP.²¹ These mechanisms offer a pathway to generate revenue without requiring explicit attribution of legal responsibility, which remains politically contested.

Additional design modifications could further strengthen the institutional framework. A funding adequacy review mechanism could require periodic assessment of financial flows relative to projected needs, creating a formal link between funding levels and loss estimates. Alternative hosting arrangements or governance reforms could also be considered to improve operational flexibility and financial autonomy. These changes do not require amendments to the Paris Agreement and can be implemented through COP decisions, making them institutionally feasible within the current legal framework.

At the same time, the findings highlight significant limitations within the existing system. Voluntary finance mechanisms are subject to free-rider dynamics, where major contributors face no enforcement consequences for non-compliance. While some donors provide substantial voluntary contributions that partially compensate for the absence of legal obligation, these flows remain inconsistent and insufficient relative to the scale of need. The reliance on loan-based finance through Multilateral Development Banks introduces further constraints, particularly for vulnerable countries already experiencing debt pressures. These factors reinforce the conclusion that the current model cannot deliver adequate funding under existing conditions.

While this study focuses on the formal UNFCCC process, it does not capture the full landscape of climate finance, including bilateral initiatives, regional mechanisms, and non-state contributions. While they may provide supplementary funding, they do not alter the central finding that the primary multilateral mechanism lacks the authority required to mobilise finance at scale. Future research should therefore examine the interaction between formal and informal funding channels to assess their combined effectiveness in addressing loss and damage.

The analysis presented here is based on institutional design prior to full operationalisation of the FRLD. Comprehensive disbursement data will only become available over time. While early voluntary contributions may partially offset structural limitations in the short term, the absence of binding mechanisms suggests that such outcomes are unlikely to be sustained. Future empirical studies should test this proposition using longitudinal data on pledges, disbursements, and funding gaps.

In conclusion, the findings demonstrate that the limitations of the FRLD are embedded in its institutional design rather than arising from temporary political conditions. The shift from liability to cooperation, reinforced through ambiguous legal language, has produced a framework that enables agreement but constrains financial capacity. Addressing the L&D funding gap therefore requires not only increased political commitment but also a reconsideration of the legal and institutional structures that govern how finance is generated. Without such changes, the gap between projected losses and available funding is likely to persist, regardless of improvements in implementation or governance.

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ANNEXURES

Table of all Financial Instruments Proposed

Year	Proposing Party	Instrument Type / Action	Status / Outcome	Description / Grounds for Dilution
1991	AOSIS / Vanuatu	International Insurance Pool	<i>Rejected / Proposed</i>	Based on mandatory contributions (GNP/\$CO ₂); dropped from UNFCCC 1992 for liability concerns.
2001	AOSIS / G77	Compensation & Rehabilitation Fund	<i>Rejected</i>	Subsumed into SCCF as an adaptation instrument; compensation element removed.
2008	AOSIS	Multi-Window Mechanism	<i>Rejected</i>	Included insurance, rehab, and risk management. Led to WIM (2013) which lacked financial authority.
2012	Bolivia, China, et al.	Solidarity Fund	<i>Proposed</i>	Joint statement for a fund to compensate for permanent and irreversible damage.
2012-19	Tuvalu	National Legislative Acts	<i>Adopted</i>	Established "Te Kaniva" and Tuvalu Survival Fund for domestic L&D response.
2013	G77 and China	Dedicated Financial Facility	<i>Rejected</i>	Decision 2/CP.19 stripped financial components; "compensation" and "liability" expunged.
2013	UNFCCC Parties	Warsaw Int. Mechanism (WIM)	<i>Adopted</i>	Established to address L&D from extreme/slow-onset events via knowledge.
2014	LDC Group	Displacement Coordination Facility	<i>Under Negotiation</i>	Proposal to assist in organized migration and relocation.
2015	UNFCCC Parties	Paris Agreement (Article 8)	<i>Adopted</i>	Anchored L&D as a pillar, but explicitly excluded a basis for liability/compensation.
2016	Saul Luciano Lliuya	Lawsuit vs. RWE (Germany)	<i>Under Negotiation</i>	Seeking risk management costs in Peru proportional to historical emissions.
2017-21	Uruguay, Nepal, et al.	National NDC Mechanisms	<i>Proposed</i>	National pledges to develop L&D assessment and agricultural insurance.
2018	Vanuatu	Climate Damages Tax (CDT)	<i>Proposed</i>	Proposal for a tax on fossil fuel companies to shift costs to responsible actors.
2018	Marshall Islands	Tile Til Eo 2050 Strategy	<i>Adopted</i>	Strategy identifying asset elevation and relocation as national responses.
2018-21	Bangladesh	National L&D Mechanisms	<i>Adopted</i>	Efforts to establish adaptation practices (rice varieties, deep tube-wells).

2019	UNFCCC Parties	Santiago Network	<i>Adopted</i>	Created at COP25 to catalyse technical assistance for vulnerable nations.
2021	G77 and China	Glasgow Loss & Damage Facility	<i>Rejected</i>	Blocked by US/EU/Australia; replaced by the non-binding "Glasgow Dialogue."
2021-22	Scotland, Wallonia	Early Financial Pledges	<i>Pledged</i>	Scotland pledged £7 million as early support for L&D funding.
2022	COP / CMA	Funding Arrangements & Fund	<i>Adopted</i>	Decision 2/CP.27: Historic agreement to establish a dedicated L&D Fund.
2022	V20 and G7	Global Shield	<i>Adopted</i>	Launch of insurance-based solutions for extreme weather events.
2023	AOSIS / LDC Group	Independent FRLD	<i>Partially Rejected</i>	Pushed for mandatory targets; the outcome was voluntary/World Bank-hosted.
2023	COP / CMA	Operationalization of FRLD	<i>Adopted</i>	Decisions 1/CP.28 and 5/CMA.5: Approved Governing Instrument for the Fund.
2023	Transitional Committee	Recommendations Mandate	<i>Mandated</i>	Tasked with designing the structure and governance of the new Fund.
2023	UNDRR and UNOPS	Santiago Network Secretariat	<i>Adopted (MOU)</i>	MOU for hosting the secretariat and financial arrangements.
2023	Pacific Islands	Pacific Resilience Facility	<i>Established</i>	Regional initiative focused on responding to L&D in the Pacific.
2024	FRLD Board	Naming & 2024-25 Workplan	<i>Adopted</i>	Official name: 'Fund for responding to Loss and Damage' (FRLD).
2024	Philippines	Host Country for FRLD Board	<i>Adopted</i>	Selected as host, Republic Act 12019 granted the Fund legal personality.
2024	World Bank	Hosting & Trustee Agreement	<i>Adopted</i>	Formalised the World Bank as interim trustee (MTO No. 069040) for 4 years.
2024	Global Donors	Initial Capitalization	<i>Pledged</i>	Various nations offered financial contributions to establish the Fund.
2024	Adaptation Fund Board	Resource Mobilization	<i>Adopted</i>	USD 300M target and a global small grants aggregator program.
2025	FRLD Board	Barbados Modalities (BIM)	<i>Adopted</i>	Start-up phase: USD 250M for 2025-26 for grant-based local interventions.
2025	FRLD Board	2026 Workplan & Resource Strat	<i>Adopted</i>	Strategy for 2027 replenishment; access via AF, GEF, and GCF entities.
2025-26	Haiti, Jamaica, et al.	National Project Proposals	<i>Under Review</i>	Proposals for flood recovery/water resilience (up to USD 20M each).
2025-26	FRLD Secretariat	Organizational Structure	<i>Provisional</i>	Development of PACE, GBA, BFM units and assessment methodologies.

Table A.1: Table of all Financial Instruments Proposed

Country-Country Matrix

Bloc	Phase I & II <i>Pre WIM (1991-2013)</i>	Phase III <i>WIM (2013-15)</i>	Phase IV <i>Paris Agreement (2015-22)</i>	Phase V <i>FRLD Design (2022-Present)</i>
AOSIS	Pushed insurance & polluter-pays framing	Pushed for a mechanism with a compensation at COP19.	Accepted Art. 8 liability exclusion to secure Paris.	Called for mandatory funds but accepted the design at COP28.
LDC Group	Focused on responsibility and restitution language.	Co-sponsored strong WIM mandate; pushed assessed contribution proposals.	Submitted assessed-contribution proposals; eventually accepted compromise.	Accepted voluntary model; prioritized operationalization over structural reform.
EU	Resisted liability and framed L&D within adaptation finance.	Insisted on liability exclusion as a condition for the WIM agreement.	Authored the exclusion language in Decision 1/CP.21; opposed dedicated fund at COP26.	Supported FRLD establishment contingent on no liability; pledged voluntary funds.
Umbrella Group	Opposed any compensation; framed all L&D as adaptation (US/Japan/Aus).	Blocked strong WIM mandates; insisted on voluntary/capacity-building frames.	US specifically excluded from Art. 8 obligations; no mandatory finance commitments.	Accepted FRLD as a voluntary mechanism; US pledged USD 17.5M.

Table A.2: Country-Coalition Position Matrix (1991–present) tracking how the language of L&D governance shifted among key negotiating blocs.

Contribution Status of FRLD

Treemap of Contribution Status

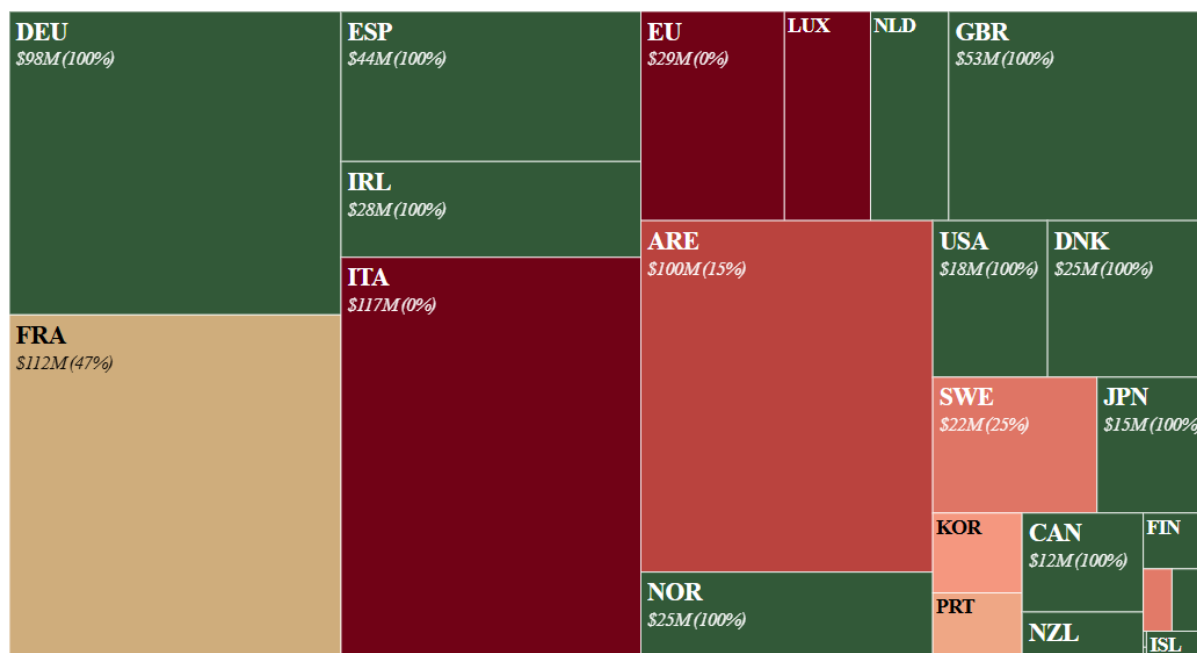


Figure A.1: FRLD Contributor Proportion Treemap. Area represents total amount pledged (USD M). Colours portray the percentage of funds actually received with green signifying a higher percentage of funds reaching FRLD compared to yellow and red. The countries are coded with ISO 3166 Alpha 3 Codes.

Detailed Table of Analysis

<i>Contributor</i>	<i>Curr</i>	<i>Pledge</i>	<i>Eff. Contribution</i>	<i>Cash Receipts</i>	<i>Promissory Notes</i>	<i>Total Received</i>	<i>Outstanding Pledges</i>	<i>Outstanding Contributions</i>
Australia	AUD	33.14	50	11.08	-	11.08	-	22.06
Austria	EUR	27.64	25	15.9	-	15.9	-	11.75
Belgium	EUR	3.4	2	2.23	-	2.23	1.17	-
Canada	CAD	11.63	16	11.63	-	11.63	-	-
Denmark	DKK	24.66	175	24.66	-	24.66	-	-
Estonia	EUR	0.05	0.05	0.05	-	0.05	-	-
European Commission	EUR	29.37	-	-	-	-	29.37	-
Finland	EUR	3.52	3	3.52	-	3.52	-	-
France	EUR	111.81	50	53.07	-	53.07	58.74	-
Germany	EUR	98.24	92	98.24	-	98.24	-	-
Iceland (USD)	USD	0.6	0.6	0.6	-	0.6	-	-
Iceland (ISK)	ISK	2.55	-	-	-	-	2.55	-
Ireland	EUR	28.1	25	28.1	-	28.1	-	-
Italy	EUR	117.47	-	-	-	-	117.47	-
Japan	USD	15	15	15	-	15	-	-
Korea	USD	7	7	2.15	-	2.15	-	4.85
Latvia	EUR	0.05	0.05	0.05	-	0.05	-	-
Luxembourg	EUR	17.62	-	-	-	-	17.62	-
Netherlands	EUR	15.78	15	15.78	-	15.78	-	-
New Zealand	NZD	5.66	10	5.66	-	5.66	-	-
Norway	USD	25	25	25	-	25	-	-
Portugal	EUR	5.69	5	2.16	-	2.16	-	3.52
Slovenia	EUR	1.73	1.5	0.44	-	0.44	-	1.29
Spain	EUR	44.01	40	44.01	-	44.01	-	-
Sweden	SEK	21.64	200	5.33	-	5.33	-	16.32
United Arab Emirates	USD	100	100	15	-	15	-	85
United Kingdom	GBP	53.26	40	26.35	26.91	53.26	-	-
United States	USD	17.56	17.56	17.56	-	17.56	-	-
Total		822.19	—	413.57	26.91	440.48	226.92	144.79

Table A.3: Pledges and Contributions as of December 31, 2025. The total amount includes USD 10 million paid by Japan to finance the FRLD Interim Secretariat through the UNFCCC prior to the Fund's establishment, as well as USD 5 million paid directly by Japan to the FRLD Trust Fund. Promissory note contributions are represented separately from paid-in cash, reflected by actual US dollar value of cash contributions or the December 31, 2025. Actual USD receipts covers both cash receipts and promissory notes. All non-USD amounts are valued based on exchange rates as of December 31, 2025. Note that totals may not add up due to rounding.

Financial Status of FRLD

Section	Category	Amount
Donor Commitments	Contributions	585.27
	Pledges	226.92
	Total Pledges & Contributions (a)	822.19
Resources Received	Cash Receipts	413.57
	Uncashed Promissory Notes	26.91
	Investment Income	16.28
	Total Resources Received	456.75
Resources Not Yet Received	Contributions Outstanding	144.79
	Pledges Outstanding	226.92
	Total Not Yet Received	371.71
Total Potential Resources (A)		828.47
Funding Decisions	Projects & Fees	-
	Administrative Budget	31.93
	Total Funding Decisions (B)	31.93
Net Potential Resources (A – B)		796.54
Funds Position	Funds Held in Trust (Unrestricted)	448.6
	└ Cash & Investments	421.69
	└ Promissory Notes	26.91
	Approved Amounts Pending Transfer	23.77
	Currency Risk Reserve	4.04
	Total Available for Board Decisions (b)	420.79

Table A.4: FRLD Trust Fund Balance Sheet Summary. Includes USD 10 million paid by Japan to finance the interim secretariat prior to the Trust Fund's establishment. USD 250 million has been earmarked by the Board (5th meeting) for the Barbados Implementation Modalities funding window, pending allocation decisions. Totals may not exactly match due to rounding.

Repositories and Other Links

This repository serves as the archive for all projects, data, and academic documents linked to the thesis "**Tracking the Evolution of the Loss and Damage Financing Architecture under the UNFCCC**".

Website: <https://github.com/HariKrishnaKR1/loss-and-damage>